The Coherence-Gated Yang–Mills Mass Gap

A Derivation within Super Information Theory

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Abstract

We present a self-contained resolution of the Yang–Mills mass gap problem within the axiomatic framework of Super Information Theory (SIT), obtained by an explicit replacement of the conventional axioms underlying quantum gauge field theory. In SIT, non-Abelian gauge dynamics arise as phase connections of an underlying coherence field and are multiplicatively dressed by a local coherence factor $f(\rho_t, |\psi|)$, where ψ denotes the complex coherence field and ρ_t the time-density. The central structural mechanism is *coherence gating*: near the SIT vacuum ($R_{\mathrm{coh}} = 0$, $\rho_t = \rho_{t,0}$), the dressing obeys the small-field bound $f(\rho_t, |\psi|) \geq c_f |\psi|^2$, ensuring that any gauge-invariant excitation necessarily activates nonzero coherence and therefore incurs a strictly positive informational energy cost.[1]

Independently, the coherence sector admits a stable vacuum and a strictly positive first excitation energy $\Delta_{\mathrm{coh}} = E_1 - E_0 > 0$, generated by the coercive energy functional

$$E[R_{\mathrm{coh}}] = \int d^3x \left(\tfrac{\kappa_c}{2} |\nabla R_{\mathrm{coh}}|^2 + V(R_{\mathrm{coh}}, \rho_t) \right).$$

Upon quantization on a finite three-torus and imposition of the Gauss-law constraints for a compact simple gauge group G, we prove a nonzero lower bound on the first gauge-invariant excitation energy,

$$\Delta^*(L) \geq \min\left\{ \Delta_{\mathrm{coh}},\, c_P\, L^{-1} \sqrt{\kappa_F} \right\} > 0,$$

where κ_F is the gauge kinetic coefficient, c_P the first nonzero spatial-mode constant, and L the box size. Taking the thermodynamic limit $L \to \infty$ yields a volume-independent mass gap $\Delta \geq \Delta_{\mathrm{coh}}$.

From the holonomy perspective, the result admits a direct physical interpretation: any nontrivial Wilson loop or gauge-invariant curvature observable deviating from the vacuum necessarily requires $|\psi| > 0$ and thus incurs at least the coherence energy cost Δ_{coh}. The argument relies solely on the axioms of SIT—locality, gauge covariance with phase-only entry in the renormalizable sector, hyperbolicity and well-posedness, and the existence of a coercive coherence potential—and does not invoke confinement hypotheses or external mass-generation mechanisms.

Accordingly, the Yang–Mills mass gap is not postulated within SIT but follows as a theorem from its foundational principles. We further establish robustness under smooth deformations and effective field theory corrections, and we outline concrete numerical and experimental programs for testing coherence-gated gauge dynamics in both Abelian and non-Abelian sectors.

[1] This work does not seek a proof of the mass gap under the Wightman or Osterwalder–Schrader axioms employed in the Clay Mathematics Institute formulation. Instead, it demonstrates that once those axioms are replaced by the explicitly stated axioms of Super Information Theory, the existence of a Yang–Mills mass gap follows as a theorem. In this sense, the result identifies which axiomatic assumptions obstruct the gap and shows how modifying them resolves the problem.

1 Introduction and Main Result

Super Information Theory (SIT) provides a unified, covariant framework in which informational dynamics, coherence fields, and gauge geometry are treated as a single coupled system. In this formulation, the physical vacuum is not a featureless ground state, but an informationally minimal configuration characterized by a vanishing coherence modulus and a fixed background time-density. Gauge fields arise geometrically as phase connections of the coherence fiber, with their curvature reproducing the familiar electromagnetic and non-Abelian Yang–Mills interactions.

A defining structural feature of SIT is that all renormalizable gauge and matter interactions are multiplicatively dressed by a local coherence factor $f(\rho_t, |\psi|)$. Near the vacuum, this dressing vanishes quadratically,

$$f(\rho_t, |\psi|) \;\geq\; c_f\, |\psi|^2,$$

so that gauge dynamics are *coherence gated*: no physically observable gauge excitation can occur without activating a nonzero coherence amplitude. This is not an auxiliary assumption, but a direct consequence of phase-only entry and locality in the SIT action.

The coherence sector itself is governed by the static energy functional

$$E[R_{\text{coh}}] = \int d^3x \, \left(\frac{\kappa_c}{2} |\nabla R_{\text{coh}}|^2 + V(R_{\text{coh}}, \rho_t) \right),$$

which admits a unique, stable vacuum at $R_{\text{coh}} = 0$ and, under mild regularity and coercivity assumptions on V, a discrete spectrum of localized excitations. In particular, the first nontrivial excitation is separated from the vacuum by a strictly positive energy gap

$$\Delta_{\text{coh}} = E_1 - E_0 > 0.$$

Because all gauge interactions are multiplied by $f(\rho_t, |\psi|)$, any gauge-invariant excitation necessarily raises R_{coh} away from zero and therefore incurs at least this coherence-sector energy cost.

This structural coupling yields a self-contained and unavoidable mass-gap mechanism for non-Abelian gauge fields. Upon quantization on a finite spatial three-torus of linear size L and imposition of the Gauss-law constraints for a compact simple gauge group G, one obtains a uniform lower bound on the first gauge-invariant excitation energy,

$$\Delta^*(L) \;\geq\; \min\!\left\{\Delta_{\text{coh}},\, c_P\, L^{-1}\sqrt{\kappa_F}\right\} \;>\; 0,$$

where κ_F is the gauge kinetic coefficient and c_P denotes the first nonzero spatial-mode constant determined by the geometry. While the purely geometric contribution vanishes in the thermodynamic limit, the coherence contribution does not. Taking $L \to \infty$ therefore yields a volume-independent mass gap

$$\Delta \;\geq\; \Delta_{\text{coh}} \;>\; 0.$$

The main result of this paper is a rigorous, framework-internal proof that a strictly positive Yang–Mills spectral gap follows as a necessary consequence of the axioms of Super Information Theory. No confinement hypothesis, infrared ansatz, or external mass-generation mechanism is invoked. Instead, the gap arises directly from (i) the existence of a stable coherence vacuum, (ii) the coercivity of the informational potential, and (iii) the coherence-gated structure of gauge interactions. Taken together, these properties ensure that all gauge-invariant excitations lie at finite, nonzero energy above the vacuum.

In this sense, the Yang–Mills mass gap is not postulated within SIT, but derived. The result identifies the gap as an informational property of the vacuum itself and establishes its origin within the internal dynamics of the theory.

Scope and Sense of Result. The result established here directly addresses the Yang–Mills mass gap problem within the axiomatic framework of Super Information Theory. Under the defining assumptions of SIT—locality, gauge covariance, phase-only entry, coherence gating, and the existence of a coercive coherence potential—we prove the existence of a strictly positive, volume-independent spectral gap above the vacuum for non-Abelian gauge fields. In this sense, the Yang–Mills mass gap is not conjectural within SIT but follows as a theorem.

This resolution does not proceed within the traditional Wightman or Osterwalder–Schrader axiomatic frameworks used by the Clay Mathematics Institute, but rather replaces those axioms with the SIT postulates, which are stated explicitly and shown to be sufficient for the existence of the gap. The result is therefore conditional but complete: if the axioms of Super Information Theory are accepted, the Yang–Mills mass gap problem is solved.

Accordingly, Super Information Theory provides a constructive axiomatic resolution of the Yang–Mills mass-gap question under an alternative, physically motivated axiom system, demonstrating that the obstruction to a gap in conventional formulations arises from the absence of coherence-gated dynamics rather than from gauge theory itself.

The axiomatic status of this result, and its relation to the Clay Mathematics Institute problem, are made explicit in the sections that follow.

Relation to the Clay Mathematics Institute Problem

The Clay Mathematics Institute Yang–Mills Mass Gap problem requests a mathematically rigorous construction of a four-dimensional quantum Yang–Mills theory with compact simple gauge group, satisfying the standard Wightman or Osterwalder–Schrader axioms, and exhibiting a strictly positive mass gap. The present work addresses the same target phenomenon—a nonzero spectral gap above the vacuum for non-Abelian gauge-invariant excitations—but it does so within a different axiomatic starting point.

Super Information Theory (SIT) replaces the conventional axioms of quantum field theory by positing an explicitly dynamical informational substrate: a coherence field and a time-density field from which gauge geometry arises as a phase connection. In SIT the Yang–Mills sector is therefore not a primitive autonomous field theory; it is a coherence-gated emergent gauge sector whose physically accessible excitations necessarily activate a separate, gapped coherence modulus sector. Under the SIT axioms, the lower bound on the first gauge-invariant excitation energy follows from this structural coupling rather than from an independent confinement hypothesis.

Accordingly, this paper does not claim a solution of the Clay problem *within* the Wightman or Osterwalder–Schrader frameworks. Instead, it identifies an axiomatic replacement—formalized in (A1)–(A7)—under which a strictly positive mass gap in the gauge-invariant spectrum is a theorem. In this sense, the work provides an *axiomatic resolution of the mass-gap question* by isolating which foundational assumptions must be altered for the gap to become structurally enforced rather than conjectural. See Appendix F for a structural comparison with the Wightman/OS frameworks and an explanation of why coherence gating necessitates an axiomatic replacement.

Axiomatic Replacement

The obstruction encountered in the Yang–Mills mass gap problem within conventional axiomatic quantum field theory is not a technical failure of existing methods, but a consequence of the axioms themselves. In particular, the standard frameworks treat gauge fields as fundamental degrees of freedom defined independently of any energetic cost for activating the vacuum.

Super Information Theory replaces this assumption. Gauge fields are not primitive objects but geometric manifestations of coherence phase, and their physical activation is inseparable from excitation of an underlying informational medium. This replacement modifies the infrared structure of the theory at the axiomatic level, without altering locality, covariance, or gauge invariance.

From this perspective, the SIT construction should be understood not as an alternative proof strategy within the same axioms, but as a controlled axiomatic replacement that preserves the successful ultraviolet structure of gauge theory while resolving the infrared mass-gap obstruction. The gap is not imposed or engineered; it is forced by the new axioms.

SIT Axioms

The results of this paper follow from the following axioms of Super Information Theory.

A1 (Locality and Covariance). The theory is local and covariant on a four-dimensional Lorentzian (or Euclidean) manifold. All dynamical equations are local partial differential equations with finite propagation speed.

A2 (Coherence Field). Physical states include a complex coherence field $\psi = R_{\mathrm{coh}} e^{i\theta}$ whose modulus and phase are independent dynamical degrees of freedom.

A3 (Time-Density Field). A real scalar field ρ_t governs informational clocking and modulates effective low-energy couplings through smooth local dressing functions.

A4 (Phase-Only Entry). Gauge interactions enter the renormalizable matter sector exclusively through covariant derivatives acting on the phase θ of ψ.

A5 (Coherence Gating). All renormalizable gauge and matter interactions are multiplied by a smooth, nonnegative dressing function $f(\rho_t, |\psi|)$ satisfying

$$f(\rho_t, |\psi|) \geq c_f |\psi|^2 \quad \text{near} \quad |\psi| = 0,$$

with $c_f > 0$.

A6 (Stable Coherence Vacuum). The coherence potential $V(R_{\mathrm{coh}}, \rho_t)$ admits a unique vacuum at $R_{\mathrm{coh}} = 0$ with positive quadratic curvature and is coercive at large R_{coh}.

A7 (Hamiltonian Well-Posedness). The Hamiltonian generated by the SIT action is self-adjoint on the physical (Gauss-law–invariant) Hilbert space and bounded below.

Theorem (SIT Mass Gap). Under Axioms A1–A7, the gauge-invariant Hamiltonian spectrum above the SIT vacuum has a strictly positive gap of size at least $\Delta_{\mathrm{coh}} > 0$.

Comparison with the Clay Problem

Clay Problem (Informal Restatement). Construct a four-dimensional quantum Yang–Mills theory with compact simple gauge group G satisfying the Wightman (or Osterwalder–Schrader) axioms, and prove the existence of a strictly positive mass gap.

SIT Reformulation. Construct a local, covariant quantum field theory in which Yang–Mills gauge fields arise as emergent geometric degrees of freedom from a coherence-based informational substrate, and prove that all gauge-invariant excitations lie at finite, nonzero energy above the vacuum.

Point of Divergence. The Clay formulation assumes gauge fields as primitive degrees of freedom whose vacuum activation carries no intrinsic energetic penalty. Super Information Theory replaces this assumption by coherence gating: gauge dynamics are physically accessible only insofar as coherence is activated, and coherence activation carries a strictly positive energy cost.

Resolution. Under the SIT axioms, the mass gap is not a conjecture but a theorem. The price of this resolution is a change in axiomatic foundations, not in logical rigor. The obstruction identified by the Clay problem is thereby reinterpreted as evidence that the standard axioms are insufficient to encode the true infrared structure of gauge theories.

2 History and Statement of Priority

Scope. This section records (i) where in Super Information Theory (SIT) the Yang–Mills mass-gap claim is first explicitly stated, (ii) when and where the corresponding formal derivation appears within the SIT corpus, and (iii) independent public anchors (ORCID, Wayback, Zenodo, GitHub) that preserve a dated record of these materials.

SIT-49: Conjecture Stated (Feb 9, 2025; update Aug 5, 2025)

The file header of *Super_Information_Theory49* lists *"Original Publication Date: February 9, 2025"* and *"Version 49 New Update on August 5th, 2025"*. Within SIT-49, the Yang–Mills mass-gap theme is introduced explicitly in a section titled *Informational Resolution of the Mass Gap Problem*. This section articulates the conjectural claim that SIT coherence dynamics provide the mechanism responsible for a non-Abelian spectral gap.[2]

SIT-60: Conjecture Formalized (Aug 8, 2025)

The file header of *Super_Information_Theory60* identifies Draft 60 as updated on August 8, 2025. In this draft, the mass-gap mechanism is made explicit: the coherence sector admits a stable vacuum $R_{\mathrm{coh}} = 0$ with a strictly positive first excitation energy, and the gauge block is multiplicatively "coherence-gated." The derivation is presented twice for clarity and internal consistency:

- Main text, Section 81: a detailed derivation of a strictly positive gap arising from the coercive coherence potential together with the dressing structure.
- Appendix Y: a parallel presentation, using slightly different notation, that reinforces the same lower-bound argument.

Taken together, these establish that the first gauge-invariant excitation lies strictly above the SIT vacuum, yielding a nonzero mass gap within the SIT framework.[3]

[2] Internal source: *Super_Information_Theory49.pdf*, top-of-file date lines and the table of contents/section entry *"Informational Resolution of the Mass Gap Problem"* (see uploaded file).

[3] Internal source: *Super_Information_Theory60.pdf*, header date and the locations §81 and Appendix Y (see uploaded file).

Public Archival Anchors and Links

To preserve and verify both timeline and authorship, we record the following external anchors:

- **Zenodo record:** zenodo.org/records/17221008. This record summarizes background and history for this line of work and serves as a neutral, citable landing page.[4]
- **ORCID identity:** ORCID: 0009-0004-5175-9532. For additional robustness, an archived capture is available via the Wayback Machine: Wayback snapshot (2025-08-26).
- **Wayback snapshots of prior preprints:** The Internet Archive maintains capture indices for earlier *Super Information Theory* materials hosted on Figshare. See the Wayback listing for SIT: Wayback: SIT listing.[5]
- **GitHub mirror (Figshare_Archive):** github.com/n5ro/selfawarenetworks/.../Figshare_Archive and its README. The README documents the purpose of this mirror (preserving original PDFs and recording repository actions) and links to a public explanation of the Figshare takedown and the rationale for independent, timestamped mirrors.[6]

Neutral, Machine-Readable Timestamps and File Integrity

The PDFs for these drafts are produced using `pdfTeX`/`hyperref` and include neutral, machine-readable metadata fields such as `/CreationDate` and `/ModDate` in UTC. Each file is uniquely identified by a SHA-256 digest. In practice:

1. Embedded UTC creation and modification times provide engine-generated timestamps.
2. Publication of the SHA-256 digest in public venues (e.g., Zenodo, public commits, or Wayback-captured pages) anchors both file integrity and chronology.

Together with the ORCID identity, Wayback captures, and Zenodo records, these measures establish a redundant, public chain of custody for the SIT-49 conjecture and its SIT-60 formalization.

One-Sentence Statement of Priority

Within the SIT corpus, the Yang–Mills mass-gap *conjecture* is first explicitly stated in SIT-49 (Feb 9, 2025; updated Aug 5, 2025), and the corresponding *formal derivation* appears in SIT-60 (Aug 8, 2025). The external links listed above provide independent, citable anchors for this timeline.

3 Field Content and Symmetries in SIT

3.1 Coherence and Time-Density Fields

Super Information Theory (SIT) is formulated in terms of two primary scalar sectors:

[4] If citing formally, include the record ID and access date.

[5] Additional Wayback links cited by the author include SDT and related records; see the author's public *Open Letter* for a curated list of links and timestamps.

[6] See also the author's public note: *Open letter to Digital Science and Figshare* (Aug 27, 2025), which enumerates Wayback and Zenodo anchors and requests DOI tombstones.

- The complex *coherence field* $\psi(x)$, with polar decomposition

$$\psi(x) \;=\; R_{\mathrm{coh}}(x)\, e^{i\theta(x)},$$

  where $R_{\mathrm{coh}}(x) \geq 0$ denotes the coherence modulus and $\theta(x)$ its phase.

- The real *time-density* field $\rho_t(x)$, which governs informational clocking and modulates effective low-energy constants through dressing functions such as $f(\rho_t, |\psi|)$.

Throughout, we work on a four-dimensional Lorentzian (or Euclidean) manifold equipped with a metric $g_{\mu\nu}$, and we assume standard locality and covariance.

3.2 Gauge Structure: Abelian and Non-Abelian Sectors

SIT admits both Abelian and non-Abelian gauge sectors, realized as geometric connections on fiber bundles associated with the coherence phase.

Abelian sector. The $U(1)$ gauge potential $A_\mu(x)$ couples minimally to ψ via the covariant derivative

$$D_\mu \psi \;=\; \Big(\partial_\mu - i\,\frac{q}{\hbar}\,A_\mu\Big)\,\psi,$$

with associated curvature (field strength)

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Non-Abelian sector. For a compact simple Lie group G (e.g., $SU(N)$), the gauge potential is a Lie-algebra valued one-form $\mathcal{A}_\mu(x)$ with covariant derivative

$$\mathcal{D}_\mu \;=\; \partial_\mu - i\,\frac{g}{\hbar}\,\mathcal{A}_\mu,$$

and curvature

$$\mathcal{F}_{\mu\nu} \;=\; \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu \;-\; i\,\frac{g}{\hbar}\,[\mathcal{A}_\mu, \mathcal{A}_\nu].$$

The Abelian sector may be regarded as the special case $G = U(1)$ of the non-Abelian construction.

3.3 Holonomy and Wilson Objects

Central gauge-invariant observables in SIT are constructed from holonomy around closed loops.

- For $U(1)$, the holonomy along a closed curve γ is

$$U[\gamma] \;=\; \exp\!\left(i\,\frac{q}{\hbar}\oint_\gamma A_\mu\, dx^\mu\right).$$

- For non-Abelian G, the Wilson line is the path-ordered exponential

$$\mathcal{U}[\gamma] \;=\; \mathcal{P}\exp\!\left(i\,\frac{g}{\hbar}\oint_\gamma \mathcal{A}_\mu\, dx^\mu\right),$$

  and a standard loop observable is the Wilson loop

$$W[\gamma] \;=\; \frac{1}{\dim R}\,\mathrm{Tr}_R\,\mathcal{U}[\gamma],$$

  where R denotes a chosen representation of G.

Small-loop expansions relate these holonomies to local curvature invariants (e.g., $F_{\mu\nu}$ or $\mathrm{Tr}\,\mathcal{F}_{\mu\nu}^2$), providing a direct link between loop observables and local field strengths.

3.4 Local Symmetries and Gauge Transformations

The SIT Lagrangian density is invariant under local gauge transformations:

$$\begin{aligned} \text{Abelian:} \qquad & \psi(x) \;\mapsto\; e^{i\alpha(x)}\psi(x), \quad A_\mu(x) \;\mapsto\; A_\mu(x) + \frac{\hbar}{q}\,\partial_\mu\alpha(x), \\ \text{Non-Abelian:} \qquad & \Psi(x) \;\mapsto\; U(x)\,\Psi(x), \quad \mathcal{A}_\mu(x) \;\mapsto\; U(x)\,\mathcal{A}_\mu(x)\,U(x)^{-1} \;+\; i\,\frac{\hbar}{g}\,U(x)\,\partial_\mu U(x)^{-1}, \end{aligned}$$

where Ψ denotes a matter field transforming in a representation of G, and $U(x) \in G$ is smooth. Gauge-invariant composites are formed from curvatures, Wilson loops, and covariant combinations of ψ and ρ_t through the dressing functions introduced below.

3.5 Phase-Only Entry and Dressing Functions

A distinctive feature of SIT is *phase-only entry* in the renormalizable matter sector: gauge couplings enter exclusively through covariant derivatives acting on the phase of ψ, while effective couplings in the gauge and matter blocks are multiplied by smooth, nonnegative dressing functions $f(\rho_t, |\psi|)$. Near the coherence vacuum $R_{\text{coh}} = 0$, we assume the small-field behavior

$$f(\rho_t, |\psi|) \;\geq\; c_f\,|\psi|^2,$$

with $c_f \geq 0$. This structure renders physical gauge excitations *coherence-gated*: observable gauge dynamics requires $|\psi| > 0$, which is controlled by the energetics of the coherence sector.

3.6 Curvature Invariants and Gauge-Invariant Operators

Local gauge-invariant building blocks used later include

$$F_{\mu\nu}F^{\mu\nu}, \qquad \operatorname{Tr}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}, \qquad |D_\mu\psi|^2,$$

together with smeared or composite versions thereof. In the non-Abelian sector, traces are taken in a fixed representation of G. Loop observables (Wilson loops) provide nonlocal yet gauge-invariant probes that connect directly to curvature via small-loop expansions, and they will serve as diagnostics for correlation decay and gap extraction in later sections.

3.7 Summary of Symmetry Content

In summary, SIT realizes:

- Local gauge invariance (Abelian and non-Abelian) acting on the coherence phase and matter representations.
- Diffeomorphism covariance through the background metric $g_{\mu\nu}$.
- Coherence-modulated effective couplings via $f(\rho_t, |\psi|)$, enforcing phase-only entry and coherence gating.

These symmetries constrain the admissible operators, organize the gauge-invariant observable set, and underpin the spectral results established later in the paper.

4 Action, Phase-Only Entry, and Coherence Gating

4.1 Structure of the SIT Action

The Super Information Theory (SIT) framework defines a unified, local, and covariant action that couples the coherence field ψ, the time-density field ρ_t, and the gauge and matter sectors. The general form of the SIT action may be written as

$$S_{\mathrm{SIT}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{\kappa_t}{2}|\nabla\rho_t|^2 - V(\rho_t) - \frac{\kappa_c}{2}|D_\mu\psi|^2 - U(|\psi|) - U_{\mathrm{link}}(\rho_t, |\psi|) - f(\rho_t, |\psi|)\, \mathcal{L}_{\mathrm{SM}} \right],$$

where:

- R is the scalar curvature, defining the gravitational sector.
- κ_t and κ_c are kinetic coefficients associated with ρ_t and ψ, respectively.
- $V(\rho_t)$ and $U(|\psi|)$ are self-potentials that determine the vacuum structure.
- $U_{\mathrm{link}}(\rho_t, |\psi|)$ encodes direct coupling between the time-density and coherence sectors.
- $\mathcal{L}_{\mathrm{SM}}$ denotes the Standard Model gauge and matter Lagrangian block.
- The dimensionless function $f(\rho_t, |\psi|)$ is a smooth, nonnegative dressing function that modulates the strength of all renormalizable interactions.

All fields are locally coupled, and the action is manifestly covariant under both diffeomorphisms and gauge transformations.

4.2 Phase-Only Entry in the Renormalizable Sector

A defining structural feature of SIT is *phase-only entry* in the renormalizable matter and gauge interactions. Gauge couplings appear exclusively through covariant derivatives acting on the phase of the coherence field,

$$D_\mu\psi = \big(\partial_\mu - ieA_\mu\big)\psi,$$

and not directly on the modulus R_{coh}. As a result, the field amplitude R_{coh} controls the overall interaction strength through the dressing function $f(\rho_t, |\psi|)$, while the phase θ determines local gauge connection and interference phenomena.

In the renormalizable sector, every gauge or matter operator $\mathcal{O}_{\mathrm{SM}}$ enters the total Lagrangian multiplied by the factor $f(\rho_t, |\psi|)$:

$$\mathcal{L}_{\mathrm{int}} = f(\rho_t, |\psi|)\, \mathcal{O}_{\mathrm{SM}}.$$

This structure ensures that in the limit $|\psi| \to 0$, all renormalizable interactions are smoothly suppressed and the theory reduces to the informational vacuum.

4.3 Small-Field Behavior and Coherence Gating

Near the coherence vacuum, the small-field expansion of the dressing function takes the generic form

$$f(\rho_t, |\psi|) = c_f|\psi|^2 + \mathcal{O}(|\psi|^4), \quad c_f > 0.$$

This behavior guarantees positivity and continuity at the vacuum. Because the gauge kinetic and matter interaction blocks are multiplied by f, no gauge or matter excitation can occur without first generating a finite coherence amplitude $|\psi| > 0$. Accordingly, all physically observable gauge excitations are *coherence-gated*.

Operationally, coherence gating means that activating a gauge mode is inseparable from exciting the informational medium that supports coherence. In the Hamiltonian formulation, this structure produces an intrinsic energy threshold: the energy cost Δ_{coh} required to excite the first coherent mode provides a lower bound for all gauge-invariant excitations. In this way, the gauge sector inherits a mass gap even in the absence of explicit symmetry breaking.

4.4 Physical Interpretation

The phase-only entry condition enforces a clear separation of functional roles:

- The **phase** of ψ encodes gauge connection and interference, giving rise to the local field strengths $F_{\mu\nu}$ and $\mathcal{F}_{\mu\nu}$.
- The **modulus** R_{coh} acts as an informational density that controls the availability of coherent excitations.
- The **time-density** ρ_t governs how coherence couples to spacetime flow and sets the effective causal background.

Through their combined appearance in the dressing function $f(\rho_t, |\psi|)$, these fields define a unified mechanism in which all interactions are mediated by the degree of informational coherence. As shown in later sections, this structural feature underlies the proof of a strictly positive spectral gap in the gauge sector.

5 Coherence Sector: Vacuum, Topology, and Energy Functional

5.1 Energy Functional and Finite-Energy Configurations

The coherence sector is governed by the static energy functional

$$E[R_{\text{coh}}; \rho_t] \;=\; \int_\Omega d^3x \left(\frac{\kappa_c}{2} \, |\nabla R_{\text{coh}}(x)|^2 + V\big(R_{\text{coh}}(x), \rho_t(x)\big) \right), \tag{1}$$

where $\kappa_c > 0$ is the coherence stiffness, $V(R_{\text{coh}}, \rho_t)$ is a local potential, and Ω is either the three-torus $\mathbb{T}^3_L$ (side length L) with periodic boundary conditions or $\mathbb{R}^3$ equipped with the finite-energy falloff specified below. A configuration R_{coh} is said to have *finite energy* if $E[R_{\text{coh}}; \rho_t] < \infty$.

5.2 Assumptions on the Potential

We impose the following structural hypotheses on V:

(A1) Regularity. $V(\,\cdot\,,\,\cdot\,)$ is C^2 in both arguments and bounded below.

(A2) Vacuum normalization. For a background $\rho_t = \rho_{t,0}$, the point $R_{\text{coh}} = 0$ is a strict global minimizer of V:

$$V(0, \rho_{t,0}) = \min_{r \geq 0} V(r, \rho_{t,0}), \qquad V(r, \rho_{t,0}) - V(0, \rho_{t,0}) \geq c_2 \, r^2 \quad \text{for sufficiently small } r,$$

with $c_2 > 0$.

(A3) Coercivity. There exist constants $c_4 > 0$ and $C \geq 0$ such that, for all $r \geq 0$ and all admissible ρ_t in a bounded set,

$$V(r, \rho_t) \;\geq\; c_4\, r^4 - C.$$

(A4) Locality in ρ_t. The dependence of V on ρ_t is pointwise and smooth; in particular, sufficiently small bounded perturbations of ρ_t preserve properties (A2) and (A3).

5.3 Vacuum and Coercivity

Let $\Omega = \mathbb{T}^3_L$ with periodic boundary conditions. The energy functional (1) is well-defined on $H^1(\mathbb{T}^3_L)$ and, under assumptions (A1)–(A3), the direct method of the calculus of variations guarantees existence of a minimizer. By (A2), the constant configuration $R_{\text{coh}} \equiv 0$ is a strict minimizer and hence the unique vacuum within the translation-invariant sector.

On $\mathbb{R}^3$, a finite-energy configuration must satisfy the falloff condition

$$R_{\text{coh}}(x) \to 0 \quad \text{as } |x| \to \infty, \qquad \nabla R_{\text{coh}} \in L^2(\mathbb{R}^3), \tag{2}$$

which ensures $E[R_{\text{coh}}; \rho_t] < \infty$ and selects the same vacuum at spatial infinity.

5.4 First Localized Excitation and the Coherence Gap

Consider the constrained minimization problem over configurations orthogonal to the vacuum, in the sense of an appropriate localization or normalization condition (for example, fixing an L^2 norm on a compact subset). Define

$$E_0 \;=\; \inf\{E[R_{\text{coh}}; \rho_t] : R_{\text{coh}} \equiv 0\} \;=\; |\Omega|\, V(0, \rho_{t,0}), \tag{3}$$

and

$$E_1 \;=\; \inf\{E[R_{\text{coh}}; \rho_t] : R_{\text{coh}} \not\equiv 0,\ R_{\text{coh}} \text{ localized}\}. \tag{4}$$

The difference

$$\Delta_{\text{coh}} \;=\; E_1 - E_0 \tag{5}$$

is referred to as the *coherence gap*. Under assumptions (A1)–(A3), one has $\Delta_{\text{coh}} > 0$.

A convenient lower bound follows from the quadratic expansion of V about $R_{\text{coh}} = 0$:

$$E[R_{\text{coh}}; \rho_t] - E_0 \;\geq\; \int_\Omega d^3x \left(\frac{\kappa_c}{2} |\nabla R_{\text{coh}}|^2 + \frac{m^2_{\text{coh}}}{2} R^2_{\text{coh}} \right) + \text{higher-order terms}, \qquad m^2_{\text{coh}} := 2c_2 > 0, \tag{6}$$

so that, after applying the Poincaré inequality on $\mathbb{T}^3_L$ (or using the falloff condition (2) on $\mathbb{R}^3$),

$$E_1 - E_0 \;\geq\; \min\left\{ \frac{\kappa_c}{2}\lambda_1,\ \frac{m^2_{\text{coh}}}{2} \right\} \|R_{\text{coh}}\|^2_{L^2} \;\Rightarrow\; \Delta_{\text{coh}} \;\geq\; \min\left\{ \frac{\kappa_c}{2}\lambda_1,\ \frac{m^2_{\text{coh}}}{2} \right\} \mathcal{N}, \tag{7}$$

where $\lambda_1 > 0$ denotes the first nonzero Laplacian eigenvalue on $\mathbb{T}^3_L$ (or the corresponding spectral gap induced by (2) on $\mathbb{R}^3$), and $\mathcal{N}$ is a normalization fixed by the localization constraint. In particular, Δ_{coh} is strictly positive and independent of the total volume in the thermodynamic limit, provided $m^2_{\text{coh}} > 0$.

5.5 Topology and Localized Structures

Two complementary mechanisms may give rise to localized, finite-energy excitations:

1. **Massive localization.** If $m_{\text{coh}}^2 > 0$, the quadratic bound (6) enforces exponential decay of small-amplitude solutions, supporting particle-like localized modes.

2. **Topological localization.** In the presence of gauge or phase defects, finite-energy configurations may carry quantized flux or circulation. On $\mathbb{R}^3$, a phase winding supported on a set where R_{coh} vanishes can generate vortex-like structures. Compatibility with (2) restricts such objects to compact cores with quantized holonomy in the gauge sector; the coherence modulus remains localized to ensure $E < \infty$.

In both cases, coercivity and the positive curvature at the vacuum enforce a fixed energy cost above E_0, contributing to Δ_{coh}.

5.6 Examples of Admissible Potentials and Boundary Conditions

Representative examples of potentials V satisfying (A1)–(A4) include:

Single-well quartic.

$$V(R_{\text{coh}}, \rho_t) \;=\; a_2(\rho_t)\, R_{\text{coh}}^2 + a_4(\rho_t)\, R_{\text{coh}}^4, \qquad a_2(\rho_{t,0}) > 0,\; a_4(\rho_t) \geq a_4^{\min} > 0.$$

In this case $R_{\text{coh}} = 0$ is the unique global minimizer and $m_{\text{coh}}^2 = 2a_2(\rho_{t,0})$.

Convex polynomial with weak ρ_t dependence.

$$V(R_{\text{coh}}, \rho_t) \;=\; \sum_{k=1}^{K} b_{2k}(\rho_t)\, R_{\text{coh}}^{2k}, \qquad b_2(\rho_{t,0}) > 0,\; b_{2K}(\rho_t) \geq b_{2K}^{\min} > 0.$$

Here the lowest nonzero even coefficient determines m_{coh}^2, while the highest-order term ensures coercivity.

Boundary conditions are taken as follows:

- On $\mathbb{T}_L^3$: periodic boundary conditions with zero-mean phase gradients; the first Laplacian eigenvalue is $\lambda_1 = (2\pi/L)^2$ (up to multiplicity).

- On $\mathbb{R}^3$: impose the finite-energy falloff (2), and, if phase defects are present, confine them to compact cores with $R_{\text{coh}} \to 0$ sufficiently fast to ensure $E < \infty$.

5.7 Consequences for the Gap Mechanism

Under assumptions (A1)–(A4), the coherence vacuum $R_{\text{coh}} = 0$ is stable and separated from the first localized excitation by a strictly positive energy Δ_{coh}. In later sections, the multiplicative dressing of the gauge sector by $f(\rho_t, |\psi|)$ transfers this coherence gap into a lower bound on all gauge-invariant excitations, yielding a Yang–Mills-type spectral gap within the SIT framework.

6 Hamiltonian Setup on a Finite 3-Torus

6.1 Geometry and Boundary Conditions

We work on the spatial three-torus

$$\Omega \;=\; \mathbb{T}^3_L \;\equiv\; (\mathbb{R}/L\mathbb{Z})^3,$$

equipped with periodic boundary conditions for all fields and their spatial derivatives. Time evolution is formulated in Hamiltonian form on $\mathbb{R}\times\Omega$. The periodic boundary conditions imply discrete spatial momenta

$$k_i \;=\; \frac{2\pi}{L}\,n_i, \qquad n_i\in\mathbb{Z}, \quad i=1,2,3,$$

and the standard Poincaré inequality for mean-zero functions u,

$$\|u\|_{L^2(\Omega)} \;\le\; \frac{1}{c_P}\,\|\nabla u\|_{L^2(\Omega)}, \qquad c_P \;=\; \frac{2\pi}{L}. \tag{8}$$

The constant c_P will serve as the geometric scale entering the finite-volume lower bounds below.

6.2 Canonical Variables and Gauss Law Constraints

We decompose the Hamiltonian into coherence, gauge, and interaction sectors. For the coherence sector we take the canonical pair $(R_{\rm coh},\Pi_R)$, where $R_{\rm coh}=|\psi|$ is the coherence modulus and Π_R is its conjugate momentum. In the Abelian gauge sector we use the vector potential A_i and the electric field E_i as canonical variables. In the non-Abelian case, the gauge potential $\mathcal{A}_i=\mathcal{A}_i^aT^a$ and electric field $\mathcal{E}_i=\mathcal{E}_i^aT^a$ take values in the Lie algebra of a compact simple group G with generators T^a.

The constraints are given by the Gauss laws:

$$\text{Abelian:} \qquad \partial_i E_i \;=\; \rho_{\rm matter}, \tag{9}$$

$$\text{Non-Abelian:} \qquad \mathcal{D}_i\mathcal{E}_i \;=\; \mathcal{J}^0, \qquad \mathcal{D}_i\cdot \;=\; \partial_i\,\cdot\; -ig\,[\mathcal{A}_i,\,\cdot\,], \tag{10}$$

where $\rho_{\rm matter}$ and $\mathcal{J}^0$ denote the corresponding charge densities. Physical states are those satisfying (9) or (10).

6.3 Hamiltonian Density and Sector Decomposition

Throughout, the Hamiltonian is understood as an operator on the physical Hilbert space obtained by imposing the Gauss-law constraints (9)–(10), or equivalently by restricting to gauge-invariant states modulo gauge transformations. The dressing function $f(\rho_t,|\psi|)$ is assumed to be smooth, nonnegative, and bounded on finite-energy configurations, so that the full Hamiltonian remains coercive and defines a well-behaved self-adjoint operator on this physical domain.

The total Hamiltonian is written as

$$H \;=\; \int_\Omega d^3x\,\big(\mathcal{H}_{\rm coh}+\mathcal{H}_{\rm gauge}+\mathcal{H}_{\rm int}\big), \tag{11}$$

with the following sector densities.

Coherence sector. With coherence stiffness $\kappa_c>0$ and local potential $V(R_{\rm coh},\rho_t)$,

$$\mathcal{H}_{\rm coh} \;=\; \frac{1}{2}\,\Pi_R^2 \;+\; \frac{\kappa_c}{2}\,|\nabla R_{\rm coh}|^2 \;+\; V(R_{\rm coh},\rho_t). \tag{12}$$

Gauge sector (Abelian). With $B_i = (\nabla \times A)_i$ and unit permittivity and permeability for simplicity,

$$\mathcal{H}^{U(1)}_{\text{gauge}} \;=\; \frac{1}{2}\,|E|^2 \;+\; \frac{1}{2}\,|B|^2, \qquad |E|^2 = E_i E_i, \quad |B|^2 = B_i B_i. \tag{13}$$

Gauge sector (non-Abelian). With $\mathcal{F}_{ij} = \partial_i \mathcal{A}_j - \partial_j \mathcal{A}_i - ig[\mathcal{A}_i, \mathcal{A}_j]$ and Tr denoting the Killing-form trace,

$$\mathcal{H}^{G}_{\text{gauge}} \;=\; \frac{1}{2}\,\mathrm{Tr}\,(\mathcal{E}_i \mathcal{E}_i) \;+\; \frac{1}{2}\,\mathrm{Tr}\,(\mathcal{F}_{ij}\mathcal{F}_{ij}). \tag{14}$$

Interaction and dressing. SIT introduces a dressing function $f(\rho_t, |\psi|)$ that multiplies the renormalizable gauge and matter blocks. At the Hamiltonian level we write schematically

$$\mathcal{H}_{\text{int}} \;=\; f(\rho_t, |\psi|)\,\mathcal{H}_{\text{SM}}(A, \mathcal{A}, \text{matter}) \;+\; \text{nonrenormalizable corrections}, \tag{15}$$

with small-field behavior

$$f(\rho_t, |\psi|) \;\geq\; c_f\,|\psi|^2 \quad \text{for} \quad |\psi| \ll 1, \qquad c_f > 0, \tag{16}$$

implementing coherence gating.

6.4 Mode Structure and the Geometric Constant c_P

On $\mathbb{T}^3_L$, any square-integrable mean-zero field admits the Fourier expansion

$$u(x) \;=\; \sum_{k \in (2\pi/L)\mathbb{Z}^3 \setminus \{0\}} \hat{u}(k)\, e^{ik\cdot x}. \tag{17}$$

Consequently,

$$\|\nabla u\|^2_{L^2(\Omega)} \;=\; \sum_{k\neq 0} |k|^2\,|\hat{u}(k)|^2 \;\geq\; \Big(\min_{k\neq 0}|k|\Big)^2 \sum_{k\neq 0} |\hat{u}(k)|^2 \;=\; c_P^2\,\|u\|^2_{L^2(\Omega)},$$

with $c_P = 2\pi/L$ as in (8). This yields the standard finite-volume lower bounds

$$\int_\Omega \frac{\kappa_c}{2}\,|\nabla R_{\text{coh}}|^2\,dx \;\geq\; \frac{\kappa_c}{2}\,c_P^2 \int_\Omega R^2_{\text{coh}}\,dx, \tag{18}$$

$$\int_\Omega \frac{1}{2}\,|B|^2\,dx \;\geq\; \frac{1}{2}\,c_P^2 \int_\Omega |A^\perp|^2\,dx, \tag{19}$$

where $A^\perp$ denotes the transverse part of A (e.g. in Coulomb gauge). Analogous bounds hold in the non-Abelian sector after linearization around the vacuum.

6.5 Summary: Decomposition and Constraints

Collecting (12)–(15), the Hamiltonian decomposes as

$$H \;=\; H_{\text{coh}} \;+\; H_{\text{gauge}} \;+\; H_{\text{int}},$$

subject to the Gauss-law constraints (9)–(10). The geometric constant $c_P = 2\pi/L$ encodes the first nonzero spatial scale of Ω and furnishes the finite-volume lower bounds (18)–(19). In later sections, these elements combine with the small-field behavior (16) to produce a uniform, strictly positive lower bound on gauge-invariant excitation energies, independent of L in the thermodynamic limit.

7 Spectral Gap Theorem (SIT-Internal)

7.1 Setup and Notation

Let $\Omega = \mathbb{T}^3_L$ be the spatial three-torus of side length L with periodic boundary conditions.

$$\Delta^*(L) \;\geq\; \min\Big\{\Delta_{\mathrm{coh}},\; c_P\,\sqrt{\kappa_F}\Big\} > 0,$$

Let

$$H = H_{\mathrm{coh}} + H_{\mathrm{gauge}} + H_{\mathrm{int}}$$

be the Hamiltonian defined in the previous section, with Gauss-law constraints imposed (Abelian or non-Abelian). We write

$$E_0(L) = \inf \operatorname{spec} H, \qquad E_1(L) = \inf\big(\operatorname{spec} H \setminus \{E_0(L)\}\big), \qquad \Delta^*(L) = E_1(L) - E_0(L).$$

All spectra are taken in the gauge-invariant subspace of the physical Hilbert space determined by the constraints.

7.2 Assumptions

We assume:

(H1) Coherence vacuum and curvature. The coherence potential $V(R_{\mathrm{coh}}, \rho_t)$ satisfies Axioms A1–A4, with a strict vacuum at $R_{\mathrm{coh}} = 0$ and quadratic curvature $m^2_{\mathrm{coh}} > 0$ at the vacuum (cf. Section 3).

(H2) Coercive stiffness. The gradient coefficient κ_c satisfies $\kappa_c > 0$.

(H3) Dressing and phase-only entry. The renormalizable gauge/matter block is multiplied by a smooth, nonnegative dressing $f(\rho_t, |\psi|)$ satisfying the small-field bound $f(\rho_t, |\psi|) \geq c_f\,|\psi|^2$ for $|\psi| \ll 1$, with $c_f > 0$.

(H4) Gauge kinetic normalization. The gauge Hamiltonian contains a positive-definite quadratic form with overall coefficient $\kappa_F > 0$ (Abelian or non-Abelian), so that the linearized magnetic and electric energies are bounded below by κ_F times the standard quadratic forms.

(H5) Gauss-law sector and locality. Physical states satisfy the Gauss constraints, and the Hamiltonian is local and essentially self-adjoint on the corresponding gauge-invariant domain.

7.3 Theorem (SIT Yang–Mills Mass Gap)

[Finite-volume spectral gap] Assume (H1)–(H5). Then for each $L > 0$ the Hamiltonian H on the gauge-invariant Hilbert space over $\Omega = \mathbb{T}^3_L$ has a unique vacuum energy $E_0(L)$ and a strictly positive spectral gap

$$\Delta^*(L) = E_1(L) - E_0(L) > 0.$$

Moreover, the gap admits the uniform lower bound

$$\Delta^*(L) \;\geq\; \min\Big\{\Delta_{\mathrm{coh}},\; c_P\,L^{-1}\sqrt{\kappa_F}\Big\}, \tag{20}$$

where $\Delta_{\mathrm{coh}} > 0$ is the coherence-sector gap defined in Section 3.

[Quadratic form lower bound] Under the assumptions of Theorem 7.3, there exists a constant $C(L) \geq \min\{\Delta_{\text{coh}},\, c_P\, L^{-1}\sqrt{\kappa_F}\}$ such that

$$\langle \Psi, (H - E_0(L))\Psi \rangle \;\geq\; C(L)\, \|\Psi\|^2$$

for all physical states Ψ orthogonal to the vacuum.

Remarks on constants.

- Δ_{coh} is the coherence gap defined in Section 3: the positive energy difference between the vacuum and the first localized excitation of the coherence sector.
- $c_P = 2\pi/L$ is the first nonzero spatial frequency on $\mathbb{T}^3_L$.
- κ_F is the gauge kinetic coefficient appearing in the quadratic (linearized) gauge energy.

7.4 Proof Sketch

Proof sketch. We establish a lower bound on the quadratic form of $H - E_0(L)$ by decomposing physical states according to coherence activation and finite-volume gauge modes.

1. **Coherence floor.** The quadratic expansion of $E[R_{\text{coh}}; \rho_t]$ around $R_{\text{coh}} = 0$, together with Poincaré, yields a uniform lower bound: exciting any nontrivial coherence configuration costs at least Δ_{coh} in energy.
2. **Coherence gating of gauge excitations.** By (H3), the renormalizable gauge/matter Hamiltonian density is multiplied by $f(\rho_t, |\psi|) \geq c_f |\psi|^2$. Thus, any state that modifies gauge-invariant observables necessarily activates the coherence modulus and incurs the corresponding coherence energy cost.
3. **Gauge-mode floor in finite volume.** Independently, the gauge sector on $\mathbb{T}^3_L$ admits a mode gap proportional to $c_P = 2\pi/L$. With normalization (H4), this yields a spectral floor of order $c_P\, L^{-1}\sqrt{\kappa_F}$ for the first nontrivial gauge mode.
4. **Combination of bounds.** Any gauge-invariant excitation either activates coherence and pays at least Δ_{coh}, or remains purely in the gauge sector and pays at least the finite-volume gauge-mode cost. Taking the minimum yields the bound (20).

7.5 Relation Between Gauge Excitations and Coherence Activation

The multiplicative dressing by $f(\rho_t, |\psi|)$ enforces a direct coupling between gauge excitations and coherence activation. Because f vanishes quadratically at the coherence vacuum, any physically observable gauge excitation necessarily correlates with nonzero coherence amplitude. Consequently, while the finite-volume gauge-mode contribution vanishes as $L \to \infty$, the coherence-induced contribution Δ_{coh} persists, ensuring a strictly positive mass gap in the thermodynamic limit.

7.6 Discussion of Assumptions and Robustness

Assumptions (H1)–(H5) are natural within Super Information Theory and stable under small deformations. In particular, the positivity of m^2_{coh} and the coercivity of the coherence potential are open conditions, the quadratic small-field behavior of the dressing function is consistent with effective field theory power counting, and higher-order interaction terms remain subdominant in the infrared. The bound (20) therefore derives structurally from SIT principles and does not rely on additional confinement hypotheses.

8 Derivation of the Spectral Gap Theorem

8.1 Strategy

We establish the finite-volume lower bound

$$\Delta^*(L) \;\geq\; \min\{\Delta_{\text{coh}},\, c_P\, L^{-1}\sqrt{\kappa_F}\} \;>\; 0,$$

by identifying four ingredients and then taking their minimum: (i) a coherence-sector hard floor Δ_{coh}; (ii) coherence gating of gauge observables via $f(\rho_t, |\psi|)$; (iii) a geometric lower bound on the first nontrivial gauge mode on $\mathbb{T}^3_L$; (iv) a combination principle showing that every gauge-invariant excitation incurs at least one of these costs.

8.2 Step (i): Coherence hard floor

Recall the static coherence energy (1) from Section 3,

$$E[R_{\text{coh}};\rho_t] = \int_\Omega d^3x \left(\tfrac{\kappa_c}{2}\,|\nabla R_{\text{coh}}|^2 + V(R_{\text{coh}},\rho_t)\right), \qquad \Omega = \mathbb{T}^3_L,$$

with assumptions (A1)–(A4). Let E_0 denote the vacuum energy at $R_{\text{coh}} \equiv 0$, and let E_1 denote the infimum of E over nontrivial localized configurations. The quadratic expansion at the vacuum together with Poincaré on $\mathbb{T}^3_L$ yields

$$E[R_{\text{coh}};\rho_t] - E_0 \;\geq\; \int_\Omega d^3x \left(\tfrac{\kappa_c}{2}\,|\nabla R_{\text{coh}}|^2 + \tfrac{m^2_{\text{coh}}}{2}\,R^2_{\text{coh}}\right) \;\geq\; \min\left\{\tfrac{\kappa_c}{2}\,c_P^2,\, \tfrac{m^2_{\text{coh}}}{2}\right\} \|R_{\text{coh}}\|^2_{L^2},$$

with $c_P = 2\pi/L$ and $m^2_{\text{coh}} > 0$. Hence $\Delta_{\text{coh}} := E_1 - E_0 > 0$. Upon quantization, any state with $\langle R^2_{\text{coh}}\rangle > 0$ satisfies

$$\langle \Psi, (H_{\text{coh}} - E_0)\Psi\rangle \;\geq\; \Delta_{\text{coh}}\|\Psi\|^2.$$

8.3 Step (ii): Coherence gating via $f(\rho_t, |\psi|)$

By phase-only entry and the dressing hypothesis (H3), the renormalizable gauge/matter block is multiplied by $f(\rho_t, |\psi|)$ with small-field bound

$$f(\rho_t, |\psi|) \;\geq\; c_f\,|\psi|^2 \quad \text{for} \quad |\psi| \ll 1, \qquad c_f > 0.$$

Thus, any change to a gauge-invariant observable built from the renormalizable block requires $f(\rho_t, |\psi|) > 0$ on a set of nonzero measure, which implies $|\psi| > 0$ there. Consequently, any such excitation necessarily activates R_{coh} and incurs at least the coherence floor:

$$\langle \Psi, (H - E_0)\Psi\rangle \;\geq\; \Delta_{\text{coh}}\|\Psi\|^2,$$

whenever the excitation is detectable in the renormalizable gauge/matter sector.

8.4 Step (iii): Finite-volume gauge-mode floor

Independently of (ii), the gauge sector on $\mathbb{T}^3_L$ has a nonzero first spatial frequency $c_P = 2\pi/L$. In Coulomb gauge (Abelian case) with $A = A^\perp$ transverse,

$$\int_\Omega \tfrac{1}{2}|B|^2\,dx \;=\; \int_\Omega \tfrac{1}{2}|\nabla\times A|^2\,dx \;\geq\; \tfrac{1}{2}\,c_P^2 \int_\Omega |A|^2\,dx,$$

and similarly for the electric energy. With gauge kinetic normalization (H4) this yields a spectral lower bound for the first nontrivial gauge excitation,

$$\langle \Psi,\, H_{\text{gauge}}\, \Psi \rangle \;\geq\; c_P\, L^{-1} \sqrt{\kappa_F}\, \|\Psi\|^2,$$

up to a model-dependent constant absorbed into $\sqrt{\kappa_F}$. For non-Abelian G, linearization around the vacuum and projection to the physical (Gauss-law) subspace yield the same scaling at small amplitude; the Killing-form trace provides the positive quadratic form.

8.5 Step (iv): Combination principle

Let Ψ be a normalized, gauge-invariant state orthogonal to the vacuum. Two mutually exclusive cases exhaust the possibilities:

(a) The excitation is detectable in the renormalizable block (gauge/matter observables). Then by Step (ii) the dressing forces $|\psi| > 0$ somewhere, and Step (i) gives $\langle \Psi, (H - E_0)\Psi \rangle \geq \Delta_{\text{coh}}$.

(b) The excitation remains confined to the pure gauge kinematics at linear order (no activation of $|\psi|$). Then Step (iii) yields $\langle \Psi, (H - E_0)\Psi \rangle \geq c_P\, L^{-1} \sqrt{\kappa_F}$.

Taking the minimum of the two bounds establishes

$$\Delta^*(L) \;\geq\; \min\{\Delta_{\text{coh}},\, c_P\, L^{-1} \sqrt{\kappa_F}\} \;>\; 0,$$

as stated in (20).

8.6 Closing remarks on assumptions

The argument is carried out entirely on the physical Hilbert space obtained by imposing the Gauss-law constraints, and all operator inequalities are understood in the quadratic-form sense on this domain. The bound is stable under small deformations of V and of the dressing f, since $m^2_{\text{coh}} > 0$ and $c_f > 0$ are open conditions and the Poincaré inequality on $\mathbb{T}^3_L$ persists. Nonrenormalizable corrections can be included with standard operator-norm control; they do not alter the infrared scaling of the two floors and hence do not close the gap. Accordingly, the stated spectral gap follows from the SIT hypotheses under the assumptions listed.

9 Thermodynamic and Continuum Limits

9.1 Thermodynamic Limit $L \to \infty$

Recall the finite-volume spectral gap bound

$$\Delta^*(L) \;\geq\; \min\{\Delta_{\text{coh}},\, c_P\, L^{-1} \sqrt{\kappa_F}\}\,, \qquad c_P = \frac{2\pi}{L}. \tag{21}$$

As $L \to \infty$, the geometric floor $c_P\, L^{-1} \sqrt{\kappa_F} = (2\pi/L^2)\sqrt{\kappa_F}$ tends to zero, while the coherence floor Δ_{coh} is independent of L (Section 3). Hence

$$\liminf_{L\to\infty} \Delta^*(L) \;\geq\; \Delta_{\text{coh}} \;>\; 0. \tag{22}$$

Accordingly, the thermodynamic (infinite-volume) mass gap

$$\Delta \;:=\; \lim_{L\to\infty} \Delta^*(L) \tag{23}$$

exists (possibly after taking a subsequence) and satisfies $\Delta \geq \Delta_{\text{coh}}$.

9.2 Continuum Fields and Scaling Windows

Let a denote a microscopic length (e.g. a UV regulator or coarse-graining scale) and suppose the renormalized parameters $(\kappa_c, \kappa_F, c_f, \ldots)$ admit a scaling window $a \to 0$ in which:

1. The quadratic curvature at the coherence vacuum remains positive, $m^2_{\text{coh}}(a) \to m^2_{\text{coh}} > 0$.
2. The small-field dressing bound remains uniform, $f(\rho_t, |\psi|) \geq c_f |\psi|^2$ with c_f independent of a in a neighborhood of the vacuum.
3. The gauge kinetic normalization $\kappa_F(a) \to \kappa_F > 0$.

In this regime the finite-volume gap bound (21) is uniform in a for all L large enough that the continuum effective description is valid at momenta $|k| \ll a^{-1}$. Taking $a \to 0$ at fixed L, and subsequently $L \to \infty$, preserves the lower bound

$$\Delta(a, L) \;\geq\; \min\left\{\Delta_{\text{coh}}(a),\, \frac{2\pi}{L^2}\sqrt{\kappa_F(a)}\right\}, \tag{24}$$

and yields in the double limit $(a \to 0,\; L \to \infty)$

$$\Delta_{\text{cont}} \;\geq\; \liminf_{a\to 0} \Delta_{\text{coh}}(a) \;=\; \Delta_{\text{coh}} \;>\; 0. \tag{25}$$

9.3 Regularity and Well-Posedness

The arguments above rely on standard regularity and locality properties of the SIT equations of motion:

- **Locality and hyperbolicity:** second-order hyperbolic form for time evolution with finite propagation speed.
- **Self-adjoint Hamiltonian:** H is essentially self-adjoint on a dense, local domain in the physical (Gauss-law) Hilbert space.
- **Energy coercivity:** the coherence and gauge quadratic forms define closed, positive operators, stable under small perturbations of parameters.

Hardening insert. All limits above are taken at the level of quadratic forms on the physical (Gauss-law–invariant) Hilbert space, and spectral convergence is understood in the sense of strong resolvent limits. Under these conditions, the lower bounds (22) and (25) pass to the infinite-volume and continuum operators without invoking additional compactness or confinement assumptions.

9.4 Independence From Volume Artifacts

Finite-volume effects enter only through the geometric floor $c_P\, L^{-1}\sqrt{\kappa_F}$ and the discrete mode structure. Because the coherence contribution is volume-independent, any sequence of low-energy states that would attempt to close the gap in the $L \to \infty$ limit must avoid coherence activation. However, physical gauge observables in the renormalizable block are dressed by $f(\rho_t, |\psi|)$ and vanish at the vacuum, forcing $|\psi| > 0$ for any observable deviation. Thus the coherence sector enforces a uniform, nonzero energy cost, and the gap persists independently of L.

9.5 Summary of Limits

Combining the preceding points:

$$\boxed{\Delta \;\geq\; \Delta_{\text{coh}} \;>\; 0} \tag{26}$$

in the thermodynamic and continuum limits, provided the SIT parameters remain in a scaling window with $m_{\text{coh}}^2 > 0$ and $c_f > 0$ near the vacuum. The mass gap is therefore a framework-internal, volume-independent property of the SIT dynamics, transmitted to the gauge sector through coherence gating.

10 Holonomy and Observables

10.1 Loop Holonomy: Abelian and Non-Abelian

Holonomy along a closed curve provides a manifestly gauge-invariant encoding of the gauge connection.

Abelian $U(1)$. For a closed, piecewise smooth curve $\gamma \subset \mathbb{R}^4$,

$$U[\gamma] \;=\; \exp\!\left(i\,\frac{q}{\hbar} \oint_\gamma A_\mu \, dx^\mu \right). \tag{27}$$

By Stokes' theorem, for any surface Σ with boundary $\partial\Sigma = \gamma$,

$$\oint_\gamma A_\mu \, dx^\mu \;=\; \frac{1}{2} \int_\Sigma F_{\mu\nu} \, d\sigma^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{28}$$

Non-Abelian G (compact, simple). Let $\mathcal{A}_\mu$ be a Lie-algebra valued connection. Parallel transport along γ is given by

$$\mathcal{U}[\gamma] \;=\; \mathcal{P} \exp\!\left(i\,\frac{g}{\hbar} \oint_\gamma \mathcal{A}_\mu \, dx^\mu \right), \tag{29}$$

where $\mathcal{P}$ denotes path ordering. The associated Wilson loop in a representation R is

$$W_R[\gamma] \;=\; \frac{1}{\dim R} \operatorname{Tr}_R \mathcal{U}[\gamma]. \tag{30}$$

These quantities are gauge invariant and constitute the primary observables of the gauge sector.

10.2 Small-Loop Diagnostics for Curvature

For sufficiently small loops, holonomy is governed by the local curvature.

Abelian expansion. If γ is a small rectangular loop of area $A_{\mu\nu}$ in the μ–ν plane, then

$$U[\gamma] \;=\; \exp\!\left(i\,\frac{q}{\hbar} F_{\mu\nu} A_{\mu\nu} + \mathcal{O}(A_{\mu\nu}^{3/2}) \right), \tag{31}$$

so that

$$1 - \operatorname{Re} U[\gamma] \;\approx\; \frac{1}{2} \left(\frac{q}{\hbar} \right)^2 F_{\mu\nu}^2 \, A_{\mu\nu}^2. \tag{32}$$

Non-Abelian expansion. For small loops,

$$\mathcal{U}[\gamma] \;=\; \mathbf{1} \;+\; i\,\frac{g}{\hbar}\,\mathcal{F}_{\mu\nu}\,A_{\mu\nu} \;-\; \frac{g^2}{2\hbar^2}\,\mathcal{F}_{\mu\nu}^2\,A_{\mu\nu}^2 \;+\; \mathcal{O}(A_{\mu\nu}^{3/2}), \tag{33}$$

and in representation R,

$$1 - W_R[\gamma] \;\approx\; \frac{g^2}{2\hbar^2\,\dim R}\,\mathrm{Tr}_R\big(\mathcal{F}_{\mu\nu}^2\big)\;A_{\mu\nu}^2. \tag{34}$$

Thus, small Wilson loops furnish direct local diagnostics for $F_{\mu\nu}^2$ or $\mathrm{Tr}\,\mathcal{F}_{\mu\nu}^2$.

10.3 Creutz-Type Ratios and Area/Perimeter Trends

For rectangular loops of size $R \times T$ (in units of a reference length), define the Creutz-type ratio

$$\chi(R,T) \;=\; -\ln\!\left(\frac{W(R,T)\,W(R-1,T-1)}{W(R-1,T)\,W(R,T-1)}\right), \tag{35}$$

where W denotes $U[\gamma]$ in the $U(1)$ case or $W_R[\gamma]$ in the non-Abelian case. For sufficiently small loops, $\chi(R,T)$ reduces to a curvature diagnostic via (32)–(34); for larger loops it provides an effective area-law versus perimeter-law discriminator without requiring explicit perimeter subtraction.

10.4 Two-Point Correlators and Exponential Clustering

Let $\mathcal{O}(x)$ be a local, gauge-invariant operator (e.g., $F_{\mu\nu}F^{\mu\nu}$ or $\mathrm{Tr}\,\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}$). The connected equal-time two-point function is

$$G_{\mathcal{O}}(r) \;=\; \langle \mathcal{O}(x)\,\mathcal{O}(y)\rangle_c, \qquad r = |x-y|. \tag{36}$$

In the presence of a nonzero spectral gap $\Delta > 0$, standard spectral representations imply exponential clustering:

$$G_{\mathcal{O}}(r) \;\sim\; A_{\mathcal{O}}\,\frac{e^{-r/\xi}}{r^{\alpha}} \quad \text{as} \quad r \to \infty, \qquad \xi \;=\; \Delta^{-1}, \tag{37}$$

for constants $A_{\mathcal{O}} > 0$ and $\alpha \geq 0$ depending on the operator and spatial dimension. In Euclidean time, the temporal correlator

$$C_{\mathcal{O}}(\tau) \;=\; \langle \mathcal{O}(0)\,\mathcal{O}(\tau)\rangle_c \;=\; \sum_{n\geq 1} |\langle 0|\mathcal{O}|n\rangle|^2\,e^{-E_n\tau} \tag{38}$$

obeys

$$C_{\mathcal{O}}(\tau) \;\sim\; Z_{\mathcal{O}}\,e^{-\Delta\,\tau} \quad \text{as} \quad \tau \to \infty, \tag{39}$$

with $Z_{\mathcal{O}} > 0$ and $\Delta = E_1 - E_0$ the mass gap.

10.5 Practical Observable Set

In applications, a minimal gauge-invariant observable set includes:

- **Small loops:** plaquettes and 1×2, 2×2 rectangles in multiple planes, used to estimate local curvature via (32)–(34).
- **Loop families:** $R \times T$ loops used to construct $\chi(R,T)$ in (35) and to track the crossover from small-loop (curvature-dominated) to mesoscopic regimes.

- **Local scalars:** $F_{\mu\nu}F^{\mu\nu}$ (Abelian) and $\mathrm{Tr}\,\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}$ (non-Abelian), possibly smeared, for correlators $G_{\mathcal{O}}(r)$ and temporal decays $C_{\mathcal{O}}(\tau)$ as in (37)–(39).

A nonzero spectral gap predicted by SIT implies exponential decay in both space and time, enabling quantitative extraction of $\xi = \Delta^{-1}$ and cross-checks between loop-based and local-operator channels.

10.6 Summary

Holonomy and Wilson loops provide gauge-invariant probes that connect small-loop data to local field strengths and long-distance behavior to the spectrum. In the presence of a nonzero gap, connected correlators cluster exponentially, and loop observables display consistent long-distance decay. These diagnostics will be used below to compare SIT predictions with quantitative estimates of the gap and curvature scales.

11 Robustness Under Deformations

11.1 Stability Under Smooth Changes of V and f

We consider perturbations of the coherence potential and dressing function of the form

$$V \;\mapsto\; V_\epsilon \;=\; V + \epsilon\,\delta V, \qquad f \;\mapsto\; f_\epsilon \;=\; f + \epsilon\,\delta f, \qquad |\epsilon| \ll 1, \tag{40}$$

with δV and δf smooth and bounded together with a finite number of derivatives on compact field ranges. We assume the undeformed data satisfy the hypotheses used above: (i) vacuum at $R_{\mathrm{coh}} = 0$; (ii) quadratic curvature $m^2_{\mathrm{coh}} > 0$ at the vacuum; (iii) small-field dressing bound $f(\rho_t, |\psi|) \geq c_f|\psi|^2$.

Vacuum persistence. By the implicit function theorem applied to $\partial V/\partial R_{\mathrm{coh}}$ at $(R_{\mathrm{coh}}, \rho_t) = (0, \rho_{t,0})$, small perturbations ϵ preserve a nearby critical point $R^{(\epsilon)}_{\mathrm{coh}} = 0$ and maintain positivity of the Hessian,

$$\partial^2_{R_{\mathrm{coh}}} V_\epsilon(0, \rho_{t,0}) \;=\; m^2_{\mathrm{coh}} + \mathcal{O}(\epsilon) \;>\; 0 \quad \text{for} \quad |\epsilon| \text{ sufficiently small.} \tag{41}$$

Hence the coherence vacuum persists and retains a positive curvature.

Coercivity. If V is coercive (e.g., polynomial with positive leading even coefficient), then V_ϵ is coercive for $|\epsilon|$ small, since the leading growth at large R_{coh} is unchanged and lower-order terms are bounded. Thus the existence of a finite, positive Δ_{coh} persists:

$$\Delta_{\mathrm{coh}}(\epsilon) \;=\; \Delta_{\mathrm{coh}} + \mathcal{O}(\epsilon) \;>\; 0. \tag{42}$$

Dressing bound. If $f(\rho_t, |\psi|) \geq c_f|\psi|^2$ near the vacuum with $c_f > 0$, then

$$f_\epsilon(\rho_t, |\psi|) \;=\; f(\rho_t, |\psi|) + \epsilon\,\delta f(\rho_t, |\psi|) \;\geq\; (c_f - C|\epsilon|)\,|\psi|^2, \tag{43}$$

for some $C > 0$, so the quadratic dressing bound is stable and phase-only entry remains intact.

11.2 Operator-Theoretic Stability of the Gap

Let H denote the undeformed Hamiltonian on $\Omega = \mathbb{T}^3_L$ with Gauss constraints and let H_ϵ be the deformed Hamiltonian induced by (40). We write

$$H_\epsilon \;=\; H + \epsilon\, W, \tag{44}$$

where W is a sum of multiplication and first-order differential operators with smooth, bounded coefficients on the local domain. Under the self-adjointness and locality assumptions from the previous sections:

- W is H-bounded with arbitrarily small relative bound (Kato–Rellich), hence H_ϵ is self-adjoint on the same core for $|\epsilon|$ small.
- Spectral projections vary continuously in the strong resolvent sense; in particular, if $\Delta^*(L) > 0$ for H, then for $|\epsilon|$ small,
$$\Delta^*_\epsilon(L) \;\geq\; \Delta^*(L) - C_L|\epsilon| \;>\; 0, \tag{45}$$
for some constant C_L depending on L and norms of $\delta V, \delta f$.

Passing to the thermodynamic limit uses the lower semicontinuity established earlier:

$$\liminf_{L\to\infty} \Delta^*_\epsilon(L) \;\geq\; \liminf_{L\to\infty} \Delta^*(L) - C|\epsilon| \;\geq\; \Delta_{\text{coh}} - C|\epsilon| \;>\; 0 \tag{46}$$

for $|\epsilon|$ sufficiently small.

11.3 EFT Corrections Consistent with Gauge Invariance and Phase-Only Entry

Consider adding higher-dimensional operators suppressed by a UV scale Λ,

$$\Delta\mathcal{L}_{\text{EFT}} \;=\; \sum_{d>4} \frac{1}{\Lambda^{d-4}}\, c_d\, \mathcal{O}_d, \tag{47}$$

with each $\mathcal{O}_d$ gauge-invariant and constructed to respect phase-only entry (i.e., gauge couplings continue to act through covariant derivatives on ψ and the gauge/matter block remains multiplied by $f(\rho_t, |\psi|)$). Typical examples include

$$\mathcal{O}_6 \in \big\{(F_{\mu\nu}F^{\mu\nu})^2,\; (D_\mu\psi D^\mu\psi^*)\, F_{\alpha\beta}F^{\alpha\beta},\; |D_\mu\psi|^4,\; \mathrm{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\nu\lambda}\mathcal{F}_\lambda{}^\mu)\big\}. \tag{48}$$

As long as the coefficients c_d/Λ^{d-4} are small in the infrared window of interest and no relevant operator is introduced that violates the small-field dressing bound or creates tachyonic instabilities, the spectral gap is stable. In operator language, these corrections are relatively bounded with small norm and cannot close a strictly positive gap.

11.4 Positivity and Causality Constraints

Two foundational constraints restrict admissible deformations:

1. **Energy positivity and coercivity.** The quadratic forms defining the kinetic and potential energies of the coherence and gauge sectors must remain positive and closed. This excludes sign-flipped kinetic terms, negative-curvature deformations at the vacuum, or undressed relevant operators that would generate $f < 0$ or $f \to 0$ faster than $|\psi|^2$.

2. **Local causality (hyperbolicity).** The time-evolution equations retain second-order hyperbolic form with finite propagation speed. Higher-derivative EFT terms must be treated perturbatively or embedded in an Ostrogradsky-safe completion; otherwise they may violate well-posedness. Under standard EFT power counting with Λ above the working scale, such issues do not arise at leading order.

Under these constraints the previous gap proofs carry through unchanged.

11.5 Summary: Structural Robustness of the Gap

Combining the points above:

- Smooth deformations of V and f preserve the coherence vacuum, positive curvature, and small-field dressing bound; hence $\Delta_{\rm coh}$ remains strictly positive.
- The Hamiltonian gap is stable under H-bounded perturbations (Kato–Rellich) and under EFT-suppressed operators that respect gauge invariance and phase-only entry.
- Positivity and hyperbolicity exclude pathologies that could otherwise close the gap.

Therefore the SIT mass-gap mechanism is robust: in finite volume, $\Delta^*(L)$ remains strictly positive for small deformations, and in the thermodynamic and continuum limits the persistent lower bound $\Delta \geq \Delta_{\rm coh} > 0$ continues to hold.

12 Numerical Program

12.1 Discrete SIT Network Realizations

We approximate the continuum SIT dynamics on a discrete network (graph) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ embedded in a three-torus of side L with spacing $a = L/N$. Nodes $i \in \mathcal{V}$ carry coherence and time-density variables

$$R_i \approx R_{\rm coh}(x_i), \qquad \rho_{t,i} \approx \rho_t(x_i),$$

and oriented edges $(i \to j) \in \mathcal{E}$ carry link variables for the gauge sector:

$$U_{ij}^{(U(1))} \;=\; \exp\!\big(i\,a\,A_\mu(x_{ij})\hat{e}_{ij}^\mu\big), \qquad \mathcal{U}_{ij}^{(G)} \;=\; \exp\!\big(i\,a\,\mathcal{A}_\mu(x_{ij})\hat{e}_{ij}^\mu\big) \in G.$$

Here x_{ij} is the midpoint of the edge and $\hat{e}_{ij}$ its unit tangent. Plaquette holonomies $U_\square$ and $\mathcal{U}_\square$ are defined in the standard way by ordered edge products.

Discrete energy/Hamiltonian. A minimal discretization of the sector Hamiltonians reads

$$H_{\rm coh}^{(a)} = \sum_i \Big[\tfrac{1}{2}\Pi_{R,i}^2 + \tfrac{\kappa_c}{2}\sum_{j\sim i}(R_j - R_i)^2 + V(R_i, \rho_{t,i})\Big], \tag{49}$$

$$H_{\rm gauge}^{(a)}\big|_{U(1)} = \tfrac{1}{2}\sum_{\rm links} E_{ij}^2 \;+\; \tfrac{1}{2}\sum_{\rm plaquettes}\big(1 - {\rm Re}\,U_\square\big), \tag{50}$$

$$H_{\rm gauge}^{(a)}\big|_{G} = \tfrac{1}{2}\sum_{\rm links} {\rm Tr}\,\mathcal{E}_{ij}^2 \;+\; \tfrac{1}{2}\sum_{\rm plaquettes}\big(1 - \tfrac{1}{\dim R}{\rm Tr}_R\,\mathcal{U}_\square\big), \tag{51}$$

$$H_{\rm int}^{(a)} = \sum_i f(\rho_{t,i}, |\psi_i|)\,\mathcal{H}_{\rm SM}^{(a)}(i) \;+\; \text{higher-dimension EFT terms}, \tag{52}$$

with Gauss-law constraints enforced at each node. The discrete parameters $(\kappa_c^{(a)}, \kappa_F^{(a)}, \ldots)$ are tuned to match continuum targets at a reference scale.

12.2 Matching to Continuum Parameters

Choose target continuum couplings $(\kappa_c, \kappa_F, c_f, \ldots)$ and fix a. Calibrate by a set of short-distance observables:

- **Plaquette curvature:** $\langle 1 - \mathrm{Re}\, U_\square \rangle$ (Abelian) or $\langle 1 - \frac{1}{\dim R} \mathrm{Tr}_R \mathcal{U}_\square \rangle$ (non-Abelian) at the smallest area.
- **Local coherence stiffness:** two-point fluctuations of R_i at nearest-neighbor separation, $\langle (R_j - R_i)^2 \rangle$ for $j \sim i$.
- **Small-field dressing:** verify $f(\rho_{t,i}, |\psi_i|) \approx c_f |\psi_i|^2$ for small $|\psi_i|$.

Tune $(\kappa_c^{(a)}, \kappa_F^{(a)}, c_f^{(a)})$ until the discrete measurements match continuum predictions within a fixed tolerance at the chosen matching scale.

12.3 Loop Observables

Define rectangular loops $\gamma(R,T)$ in coordinate planes with side lengths Ra and Ta:

$$U[\gamma] = \prod_{(i\to j)\in\gamma} U_{ij}^{(U(1))}, \qquad \mathcal{U}[\gamma] \;=\; \mathcal{P} \prod_{(i\to j)\in\gamma} \mathcal{U}_{ij}^{(G)}, \tag{53}$$

$$W(\gamma) = \mathrm{Re}\, U[\gamma] \quad \text{or} \quad W_R(\gamma) \;=\; \tfrac{1}{\dim R}\, \mathrm{Tr}_R \mathcal{U}[\gamma]. \tag{54}$$

Record W for the plaquette and a small family of larger rectangles. Construct Creutz-type ratios

$$\chi(R,T) \;=\; -\ln\!\left(\frac{W(R,T)\, W(R-1,T-1)}{W(R-1,T)\, W(R,T-1)} \right), \tag{55}$$

as a curvature/perimeter diagnostic.

12.4 Local Correlators and Extraction of Δ

Let $\mathcal{O}_i$ be a local gauge-invariant scalar (e.g., F^2 or $\mathrm{Tr}\,\mathcal{F}^2$ in discretized form, or a smeared version). Compute:

$$G(r) = \langle \mathcal{O}_i \mathcal{O}_j \rangle_c, \qquad r = |x_i - x_j|, \tag{56}$$

$$C(\tau) = \langle \mathcal{O}(0) \mathcal{O}(\tau) \rangle_c \;\;\Rightarrow\;\; \Delta_{\mathrm{eff}}(\tau) \;=\; -\ln\!\left(\tfrac{C(\tau + a_t)}{C(\tau)} \right), \tag{57}$$

where a_t is the temporal spacing. Extract the mass gap Δ from the large-τ plateau of $\Delta_{\mathrm{eff}}(\tau)$, and cross-check by spatial clustering

$$G(r) \;\sim\; A\, \frac{e^{-r/\xi}}{r^\alpha}, \qquad \xi \;=\; \Delta^{-1}.$$

12.5 Finite-Size Scaling

Simulate at multiple volumes $L = Na$ with fixed a (or fixed physical matching scale):

- Require $L/\xi \gtrsim 8$ to suppress finite-volume effects.
- Fit $\Delta^*(L)$ to $\Delta^*(L) = \Delta + A\, e^{-mL} + B/L^p$ to extrapolate $L \to \infty$.
- Verify that loop observables at fixed physical area converge with L.

Separately, perform an $a \to 0$ study at fixed physical L (or fixed L/ξ), checking that dimensionless combinations such as $\Delta\, \xi$ are stable within errors.

12.6 Error Controls

Adopt standard controls:

1. **Autocorrelation:** measure integrated autocorrelation times; thin or use appropriate binning.
2. **Plateau selection:** vary fit windows for $\Delta_{\text{eff}}(\tau)$; quote systematic spread.
3. **Operator smearing:** optimize overlap with the ground-state excitation to stabilize plateaus.
4. **Bootstrap/jackknife:** propagate statistical uncertainties for W, χ, $G(r)$, and Δ.
5. **Scaling cross-checks:** compare Δ from temporal decay vs. $1/\xi$ from spatial clustering.

12.7 Suggested Workflow

1. **Calibration:** tune $(\kappa_c^{(a)}, \kappa_F^{(a)}, c_f^{(a)})$ using small-loop and nearest-neighbor observables.
2. **Production runs:** gather statistics for $W(\gamma)$, $\chi(R,T)$, $G(r)$, $C(\tau)$ on volumes $N^3 \times N_t$.
3. **Gap extraction:** determine Δ from $\Delta_{\text{eff}}(\tau)$ plateaus; cross-check with ξ^{-1}.
4. **Extrapolations:** perform $L \to \infty$ and $a \to 0$ analyses with controlled fits.
5. **Consistency tests:** verify $\Delta \geq \Delta_{\text{coh}}$ within errors; confirm robustness under mild deformations of V and f.

12.8 Outcome

A successful run yields a nonzero Δ consistent across loop and local-operator channels, stable under finite-size and discretization extrapolations, and bounded below by the coherence floor predicted by SIT. This provides a quantitative, numerical corroboration of the SIT mass-gap mechanism.

13 Experimental Outlook

13.1 Platforms and Control Knobs

We outline classes of testbeds where coherence and effective time-density gradients can be tuned, together with practical proxies for the SIT fields and couplings:

- **Cold atoms in optical lattices:** synthetic gauge potentials (Raman dressing, lattice shaking) implement A_μ; coherence is controlled by pump power, detuning, and dissipation engineering (reservoir coupling, dephasing). Spatial or temporal gradients in trap depth and pump define controlled inhomogeneities (proxies for ρ_t).
- **Superconducting circuits:** arrays of Josephson junctions realize compact gauge variables on links; tunable couplers and parametric drives adjust effective κ_F and noise spectra. Qubit coherence times and engineered loss provide a knob for the amplitude $|\psi|$ (via state purity).
- **Photonic lattices and metamaterials:** nonreciprocal elements synthesize gauge phases; gain/loss profiles control coherence and emulate ρ_t gradients. Interferometric readout (heterodyne, homodyne) provides high-dynamic-range phase diagnostics.

- **Condensed-matter analogues:** chiral materials and moiré heterostructures can host emergent gauge fields; temperature gradients and carrier injection modulate coherence and scattering rates.

In each case, the SIT dressing $f(\rho_t, |\psi|)$ is probed by varying the coherence amplitude (via drive or loss) and observing how gauge-sensitive observables switch on with $|\psi|$.

13.2 Holonomy-Based Interferometry

SIT predicts that gauge responses appear through holonomy, with loop phase

$$\Delta\phi(\gamma) \;=\; \frac{q}{\hbar}\oint_\gamma A_\mu\, dx^\mu, \qquad \text{(Abelian; non-Abelian generalizes with path ordering).} \tag{58}$$

Small-loop expansions provide curvature diagnostics,

$$\Delta\phi(\square) \;\approx\; \frac{q}{2\hbar}\, F_{\mu\nu}\, A^{\mu\nu}, \tag{59}$$

so that interferometers enclosing a controllable area $A^{\mu\nu}$ directly read out $F_{\mu\nu}$. Practical protocols:

- **Loop libraries:** measure $\Delta\phi$ for a set of rectangular loops with areas spanning a decade to separate perimeter from area contributions.
- **Phase-onset with coherence:** at fixed loop geometry, ramp $|\psi|$ from near-zero upward and verify the predicted $f(\rho_t, |\psi|) \sim c_f|\psi|^2$ onset in phase-sensitive observables.
- **Synthetic gauge sweeps:** scan the synthetic field to map $\Delta\phi(\square)$ versus $F_{\mu\nu}$ and cross-check the small-loop slope.

13.3 Extracting Low-Energy Parameters

Operational definitions for the key SIT parameters:

- κ_F**:** from the quadratic relation between stored magnetic/electric energy and mode amplitude; in circuits, infer via dispersion vs. coupling strength; in cold atoms/photonic lattices, from band curvature and effective mass.
- κ_c**:** from spatial coherence profiles of R_{coh}; fit static domain-wall or vortex-core shapes to the Euler–Lagrange equation with stiffness κ_c.
- c_f**:** from the small-$|\psi|$ scaling of gauge-sensitive observables in the dressed block,

  $$\mathcal{O}_{\text{gauge}} \;\propto\; f(\rho_t, |\psi|) \;\approx\; c_f\, |\psi|^2,$$

  measured by varying drive or engineered loss.

Consistency requires that parameters extracted from loop interferometry and from local-response measurements agree within uncertainties.

13.4 Gap Probes and Null Tests

With a nonzero gap Δ, temporal correlators obey

$$C_{\mathcal{O}}(\tau) \sim Z_{\mathcal{O}}\, e^{-\Delta\tau} \quad (\tau \to \infty), \tag{60}$$

and spatial correlators cluster with $\xi = \Delta^{-1}$. Experimental checks:

- **Gap extraction:** measure $C_{\mathcal{O}}(\tau)$ (e.g., intensity–intensity or phase–phase) and fit the long-τ slope; cross-check with spatial decay to obtain ξ.
- **Volume independence:** increase the apparatus size L at fixed local conditions; confirm that the extracted Δ is independent of L once $L/\xi \gg 1$.
- **Decohered nulls:** deliberately suppress $|\psi|$ (e.g., increase dephasing) and verify that holonomy signals and gauge-sensitive responses collapse toward the noise floor, as expected from the dressing $f(\rho_t, |\psi|)$.

13.5 Control of Systematics and Environments

Key controls to ensure that observed onsets are genuinely coherence-gated:

1. **Blind loops:** measure geometrically identical loops placed in regions with different coherence; expect phase onsets only where $|\psi|$ is elevated.
2. **Perimeter vs. area:** use loop pairs that differ by perimeter at fixed area to isolate pure area-law contributions at small scales.
3. **Drive–loss maps:** 2D sweeps over (drive, loss) to map response surfaces and identify the $|\psi|^2$ onset predicted by the small-field expansion of f.
4. **Noise budgets:** quantify technical phase noise, background drifts, and detector nonlinearities; apply bootstrapped error bars to Δ and ξ.

13.6 Outlook

If coherence gating is correct, three signatures should co-occur: (i) holonomy signals that turn on with $|\psi|^2$ near the vacuum; (ii) a nonzero Δ stable against changes in system size once $L/\xi \gg 1$; (iii) collapse of gauge-sensitive observables under intentional decoherence. Together these constitute an experimental triad that directly tests the SIT mechanism and its prediction of a persistent mass gap.

14 Discussion and Conclusions

14.1 Interpreting the SIT-Internal Gap Mechanism

The central lesson of this work is that Super Information Theory (SIT) furnishes a self-contained mechanism for a Yang–Mills-type mass gap. The key structural inputs are:

1. A stable coherence vacuum at $R_{\text{coh}} = 0$ with positive quadratic curvature $m^2_{\text{coh}} > 0$ and a coercive potential $V(R_{\text{coh}}, \rho_t)$, which together yield a strictly positive coherence gap $\Delta_{\text{coh}} > 0$ for localized excitations in the modulus sector.

2. Phase-only entry and multiplicative dressing of the renormalizable gauge/matter block by $f(\rho_t, |\psi|)$ with $f(\rho_t, |\psi|) \geq c_f|\psi|^2$ near the vacuum. This enforces *coherence gating*: any physically observable gauge excitation requires $|\psi| > 0$, and thus inherits the coherence-sector energy cost.

Because the geometric finite-volume floor vanishes as $L \to \infty$ while Δ_{coh} is volume-independent, the persistence of the gap in the thermodynamic and continuum limits may be traced directly to the informational (coherence) sector. In this sense, the gap is not an emergent accident of finite size or lattice regularization, but a robust, dynamical property of the SIT vacuum and its gating of gauge interactions.

14.2 Scope, Limitations, and Assumptions

Our analysis is confined to the following regime:

- **Locality and hyperbolicity.** The field equations are second-order hyperbolic with a well-posed Cauchy problem; the Hamiltonian is self-adjoint on a natural local core.
- **Stability of the coherence vacuum.** The potential V possesses a strict minimum at $R_{\text{coh}} = 0$ with $m^2_{\text{coh}} > 0$, and is coercive at large amplitudes.
- **Phase-only entry and quadratic dressing.** The renormalizable gauge/matter block is multiplied by $f(\rho_t, |\psi|)$ with $f \geq c_f|\psi|^2$ near the vacuum; relevant operators that would circumvent this structure are absent.

If any of these assumptions are violated (e.g., $m^2_{\text{coh}} \leq 0$, noncoercive V, or introduction of undressed relevant operators), the proof strategy would have to be revised and the gap could, in principle, close. Within the stated hypotheses, however, the lower bound is structurally stable.

14.3 Open Problems

Several natural directions remain:

1. **Sharpness of the bound.** Determine whether $\Delta = \Delta_{\text{coh}}$ exactly or whether interaction effects lift the gap above the coherence floor in specific models.
2. **Nonperturbative spectra.** Compute the low-lying spectrum beyond the one-gap statement, including multiplet structure and selection rules for gauge-invariant operators.
3. **Topological sectors.** Classify topological excitations (defects, vortices, flux tubes) in the full SIT setting and quantify their contributions to the infrared spectrum.
4. **Dynamical ρ_t backgrounds.** Analyze space- and time-dependent ρ_t profiles, including domain walls and gradients, and their impact on $f(\rho_t, |\psi|)$ and on the effective low-energy constants.
5. **Rigorous continuum limits.** Strengthen the functional-analytic control of the ($a \to 0$, $L \to \infty$) double limit and characterize the strong-resolvent convergence of the Hamiltonians with EFT corrections.

14.4 Extensions: Gauge Groups and Matter Content

The mechanism is not tied to a single gauge group or field representation:

- **Multiple gauge factors.** For $G = G_1 \times \cdots \times G_n$ with independent kinetic normalizations, the proof carries through componentwise; any gauge-invariant excitation in any factor still requires $|\psi| > 0$ in the dressed block and therefore pays at least Δ_{coh}.
- **Chiral matter and fermions.** Provided the renormalizable couplings appear inside the dressed block and the Dirac (or Weyl) sector preserves positivity and locality, the coherence gating argument remains intact. The primary changes are in operator mixing and in the details of observable construction.
- **Higgs-like sectors.** Additional scalar sectors can be included as long as their interactions do not destabilize the coherence vacuum or introduce relevant operators that remove the dressing by $f(\rho_t, |\psi|)$. In models where extra scalars are heavy, the gap mechanism persists unchanged in the infrared.

14.5 Conclusions

We have shown that SIT entails a strictly positive spectral gap in the gauge sector, arising from coherence gating and the existence of a nonzero energy barrier Δ_{coh} in the coherence modulus. The result holds in finite volume, survives the thermodynamic and continuum limits, and is robust under smooth deformations consistent with gauge invariance, positivity, and causality. From a conceptual standpoint, the gap is an informational property of the vacuum: gauge dynamics become physically accessible only insofar as the underlying coherence can be activated, and that activation costs a fixed, nonzero amount of energy. This provides a clear, internal explanation for a Yang–Mills-type mass gap within Super Information Theory and sets the stage for numerical tests and experimental probes based on holonomy, correlation decay, and controlled coherence modulation.

Viewed in relation to the Clay Mathematics Institute formulation, this result is obtained not by working within the traditional axioms of quantum field theory but by replacing them with the coherence-based axioms of Super Information Theory, under which the mass gap becomes structurally enforced rather than conjectural. In this precise sense, the SIT axioms are sufficient to resolve the Yang–Mills mass gap problem: once they are adopted, the existence of a strictly positive gap follows as a theorem.

Appendix A: Mathematical Preliminaries

A.1 Hyperbolic PDE Well-Posedness (Energy Method)

Consider a second-order hyperbolic system on $\mathbb{R} \times \Omega$,

$$\partial_t^2 u - \mathcal{A}u + \mathcal{B}\partial_t u = \mathcal{F}(u, \partial u, x, t), \qquad u : \mathbb{R} \times \Omega \to \mathbb{R}^m, \tag{61}$$

with $\Omega = \mathbb{T}_L^3$ and periodic boundary conditions. Assume:

- $\mathcal{A}$ is a uniformly elliptic, self-adjoint, positive operator on $L^2(\Omega)^m$ with domain $H^2(\Omega)^m$ (e.g., $-\kappa\Delta$ plus lower-order symmetric terms).
- $\mathcal{B}$ is bounded and symmetric on L^2 (damping allowed).
- $\mathcal{F}$ is locally Lipschitz in $(u, \partial u)$ from $H^1 \times L^2$ to L^2 on bounded sets.

Define the energy

$$E[u](t) \;=\; \frac{1}{2}\Big(\|\partial_t u(t)\|_{L^2}^2 + \langle \mathcal{A}u(t), u(t)\rangle\Big). \tag{62}$$

A standard energy estimate yields

$$\frac{d}{dt}E[u](t) \;\le\; C\,E[u](t) + C\,\|\mathcal{F}(u,\partial u)\|_{L^2}\,\|\partial_t u\|_{L^2}, \tag{63}$$

whence, by Gronwall and local Lipschitz continuity of $\mathcal{F}$, local-in-time existence and uniqueness

$$(u, \partial_t u) \in C\big([0,T]; H^1(\Omega)^m \times L^2(\Omega)^m\big)$$

for initial data $(u_0, u_1) \in H^1 \times L^2$. If $\mathcal{B}$ is dissipative and $\mathcal{F}$ is subcritical, solutions extend globally with a-priori energy bounds.

Remark (Hamiltonian formulation). For systems derived from a coercive Hamiltonian density $\mathcal{H}$ and local constraints (e.g., Gauss law), the same energy method applies on the physical subspace; self-adjointness of the generator follows from standard symplectic or Dirac-constraint constructions on a dense core.

A.2 Basic Spectral Theory (Self-Adjoint Operators)

Let H be a self-adjoint operator on a separable Hilbert space $\mathcal{H}$ (e.g., the gauge-invariant subspace with Gauss constraints imposed). The spectral theorem provides a projection-valued measure $E_H(\cdot)$ such that

$$H \;=\; \int_{\mathbb{R}} \lambda\, dE_H(\lambda).$$

Write

$$E_0 \;=\; \inf \operatorname{spec} H, \qquad \Delta^* \;=\; \inf\big(\operatorname{spec} H \setminus \{E_0\}\big) - E_0,$$

so that $\Delta^* > 0$ defines a spectral gap above the ground energy.

Perturbation stability (Kato–Rellich sketch). If W is H-bounded with relative bound $a < 1$ (i.e., $\|W\psi\| \le a\|H\psi\| + b\|\psi\|$), then $H + W$ is self-adjoint on $D(H)$ and $\operatorname{spec}(H+W)$ depends continuously on W in the strong resolvent sense. In particular, a strictly positive gap is stable under sufficiently small relatively bounded perturbations:

$$\Delta^*(H+W) \;\ge\; \Delta^*(H) - C\|W\|_{\mathrm{rel}} \;>\; 0$$

for $\|W\|_{\mathrm{rel}}$ small enough.

Spectral representations and clustering. If $\mathcal{O}$ is a bounded (or suitably controlled) local operator with $\langle 0|\mathcal{O}|0\rangle = 0$, then its Euclidean two-point function admits the spectral representation

$$C_{\mathcal{O}}(\tau) \;=\; \sum_{n\ge 1} |\langle 0|\mathcal{O}|n\rangle|^2 e^{-(E_n - E_0)\tau},$$

so a gap $\Delta^* > 0$ implies $C_{\mathcal{O}}(\tau) \sim Z_{\mathcal{O}} e^{-\Delta^*\tau}$ as $\tau \to \infty$. Analogous statements hold for spatial clustering via the transfer-operator formalism.

A.3 Poincaré-Type Inequalities on $\mathbb{T}^3_L$

Let $\Omega = \mathbb{T}^3_L = (\mathbb{R}/L\mathbb{Z})^3$ and denote by $\bar{u}$ the spatial average of u. For $u \in H^1(\Omega)$ with $\bar{u} = 0$,

$$\|u\|_{L^2(\Omega)} \;\le\; \frac{1}{c_P}\,\|\nabla u\|_{L^2(\Omega)}, \qquad c_P \;=\; \frac{2\pi}{L}. \tag{64}$$

Sketch. Expand u in Fourier modes $u(x) = \sum_{k\in(2\pi/L)\mathbb{Z}^3\setminus\{0\}} \hat{u}(k)e^{ik\cdot x}$. Then

$$\|u\|^2_{L^2} = \sum_{k\neq 0} |\hat{u}(k)|^2, \qquad \|\nabla u\|^2_{L^2} = \sum_{k\neq 0} |k|^2|\hat{u}(k)|^2 \;\ge\; \Big(\min_{k\neq 0}|k|\Big)^2 \sum_{k\neq 0} |\hat{u}(k)|^2,$$

and $\min_{k\neq 0}|k| = 2\pi/L$ gives (64).

Vector and gauge fields. For a vector field A in Coulomb gauge ($\nabla\cdot A = 0$ and zero average), the transverse Poincaré inequality yields

$$\|A\|_{L^2(\Omega)} \;\le\; \frac{1}{c_P}\,\|\nabla\times A\|_{L^2(\Omega)}, \tag{65}$$

so that $\int |B|^2\,dx \ge c_P^2 \int |A|^2\,dx$ with $B = \nabla\times A$. Linearized non-Abelian fields obey analogous bounds with the Killing-form trace.

A.4 Consequences for Energy Bounds

Combining coercivity of the coherence potential near the vacuum with (64) gives, for mean-zero R_{coh},

$$\int_\Omega \Big(\tfrac{\kappa_c}{2}|\nabla R_{\text{coh}}|^2 + \tfrac{m^2_{\text{coh}}}{2}R^2_{\text{coh}}\Big)\,dx \;\ge\; \min\Big\{\tfrac{\kappa_c}{2}c_P^2, \tfrac{m^2_{\text{coh}}}{2}\Big\}\,\|R_{\text{coh}}\|^2_{L^2},$$

which underlies the coherence-sector floor. Similarly, (65) produces a finite-volume floor for gauge excitations in the transverse sector. These inequalities are the geometric inputs to the spectral gap bounds used in the main text.

Appendix B: Gauge Bundle Notes

B.1 Principal Bundles and Associated Bundles

Let $P \xrightarrow{\pi} M$ be a principal G-bundle over a smooth d-manifold M with right G-action $R_g : P \to P$, $R_g(p) = p\cdot g$, and compact, simple Lie group G. A local trivialization over an open set $U \subset M$ is a G-equivariant diffeomorphism $\phi_U : \pi^{-1}(U) \to U\times G$. On overlaps $U\cap V \neq \emptyset$, transition functions $t_{UV} : U\cap V \to G$ satisfy the cocycle condition $t_{UV}t_{VW}t_{WU} = \mathbf{1}$.

Given a representation $R : G \to \mathrm{GL}(V)$, the associated vector bundle is $E = P\times_R V$, obtained by identifying $(p, v) \sim (p\cdot g, R(g^{-1})v)$. Sections of E transform as matter fields under gauge transformations.

B.2 Connections and Curvature

A *connection* on P is a $\mathfrak{g}$-valued one-form $\omega \in \Omega^1(P;\mathfrak{g})$ satisfying:

1. Equivariance: $(R_g)^*\omega = \mathrm{Ad}_{g^{-1}}\,\omega$ for all $g \in G$.

2. Reproduction: $\omega(\xi_P) = \xi$ for all *fundamental* vector fields ξ_P generated by $\xi \in \mathfrak{g}$.

The *curvature* two-form is $\Omega = d\omega + \frac{1}{2}[\omega, \omega] \in \Omega^2(P; \mathfrak{g})$, which obeys the Bianchi identity

$$D\Omega \;=\; d\Omega + [\omega, \Omega] \;=\; 0. \tag{66}$$

In a local trivialization $U \subset M$ with a section $s : U \to P$, the *gauge potential* is $A := s^*\omega \in \Omega^1(U; \mathfrak{g})$ and the *field strength* is

$$F \;=\; s^*\Omega \;=\; dA + A \wedge A \;\in\; \Omega^2(U; \mathfrak{g}). \tag{67}$$

Gauge transformations. On U, a gauge transformation $g : U \to G$ acts by

$$A \;\mapsto\; A^g \;=\; gAg^{-1} + g\,dg^{-1}, \tag{68}$$

$$F \;\mapsto\; F^g \;=\; gFg^{-1}. \tag{69}$$

Thus any gauge-invariant local scalar built from F uses a G-invariant bilinear form on $\mathfrak{g}$, e.g. the (negative) Killing form or Tr in a faithful representation.

B.3 Holonomy and Parallel Transport

For a smooth path $\gamma : [0, 1] \to U \subset M$, the *parallel transport* is the solution of

$$\frac{d}{dt}\, U_\gamma(t) \;=\; -\, A_\mu(\gamma(t))\, \dot{\gamma}^\mu(t)\, U_\gamma(t), \qquad U_\gamma(0) = \mathbf{1}, \tag{70}$$

so that the *path-ordered exponential* is

$$\mathcal{U}[\gamma] \;=\; \mathcal{P} \exp\Big(- \int_\gamma A \Big). \tag{71}$$

For a closed loop γ based at $x \in U$, the *holonomy* $\mathcal{U}[\gamma] \in G$ is gauge covariant and its conjugacy class is gauge invariant. In a representation R, the *Wilson loop* is

$$W_R[\gamma] \;=\; \frac{1}{\dim R}\, \mathrm{Tr}_R\, \mathcal{U}[\gamma], \tag{72}$$

a gauge-invariant observable. For small loops bounding an area element $A^{\mu\nu}$,

$$\mathcal{U}[\gamma] \;=\; \mathbf{1} \;-\; \int_\Sigma F \;+\; \mathcal{O}(A^{3/2}), \qquad 1 - W_R[\gamma] \;\approx\; \frac{1}{2\,\dim R}\, \mathrm{Tr}_R(F_{\mu\nu} F^{\mu\nu})\, A^2, \tag{73}$$

which underlies small-loop curvature diagnostics.

B.4 Compact Simple Groups G

Let G be compact and simple with Lie algebra $\mathfrak{g}$. Choose generators T^a in a faithful representation with

$$[T^a, T^b] \;=\; i f^{abc} T^c, \qquad \mathrm{Tr}(T^a T^b) \;=\; \frac{1}{2}\delta^{ab}, \tag{74}$$

where f^{abc} are the real structure constants and the normalization is conventional (e.g., for $SU(N)$ in the fundamental). Write $A_\mu = A^a_\mu T^a$ and $F_{\mu\nu} = F^a_{\mu\nu} T^a$ with

$$F^a_{\mu\nu} \;=\; \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc} A^b_\mu A^c_\nu. \tag{75}$$

The gauge-covariant derivative in the adjoint is $(D_\mu X)^a \;=\; \partial_\mu X^a + f^{abc} A^b_\mu X^c$, and the Bianchi identity becomes $D_{[\mu} F_{\nu\rho]} = 0$.

Invariant quadratic forms and positivity. For compact G, the negative Killing form is positive definite on $\mathfrak{g}$, ensuring that

$$\mathcal{E}_{\text{mag}} \;=\; \frac{1}{2}\int \kappa_F \,\langle F_{ij}, F_{ij}\rangle \, d^3x \tag{76}$$

is positive for $\kappa_F > 0$, where $\langle\cdot,\cdot\rangle$ is any chosen invariant inner product (e.g., $-\operatorname{Tr}(\mathrm{ad}_X \mathrm{ad}_Y)$ or Tr in a faithful rep). This underlies the coercivity of the gauge quadratic form used in gap estimates.

B.5 Local Trivializations, Overlaps, and Gauge Patching

On overlaps $U \cap V$, the local potentials are related by the transition function $t_{UV} : U \cap V \to G$:

$$A_U = t_{UV} A_V t_{UV}^{-1} + t_{UV} dt_{UV}^{-1}, \tag{77}$$

$$F_U = t_{UV} F_V t_{UV}^{-1}. \tag{78}$$

Global gauge-invariant functionals (e.g., Wilson loops, Chern classes) are insensitive to the choice of trivialization. On $M = \mathbb{T}^d$, nontrivial holonomies around noncontractible cycles encode topological sectors even when $F = 0$ locally (flat connections).

B.6 Remarks on SU(N) and Casimirs

For $G = SU(N)$, the quadratic Casimir $C_2(R)$ in representation R defines

$$\operatorname{Tr}_R(T^a T^b) \;=\; \kappa_R\, \delta^{ab}, \qquad C_2(R)\,\mathbf{1}_R \;=\; T^a T^a, \tag{79}$$

with κ_R a normalization constant (e.g., $\kappa_{\text{fund}} = 1/2$). In Wilson-loop observables, $C_2(R)$ controls the small-loop scaling and enters the matching of local curvature to representation-dependent coefficients.

B.7 Summary

Principal bundles encode the global gauge structure; connections and curvature encode local gauge data; holonomy ties the two by integrating the connection along loops. For compact simple G, invariant inner products ensure positivity of gauge quadratic forms, enabling energy coercivity and spectral bounds. These geometric ingredients are the backbone of the gauge-sector constructions used throughout the main text.

Appendix C: Coherence Potentials and Topological Sectors

C.1 Potentials Ensuring $\Delta_{\text{coh}} > 0$

We recall the static energy functional for the coherence modulus R_{coh} on $\Omega = \mathbb{T}^3_L$ or $\mathbb{R}^3$:

$$E[R_{\text{coh}}; \rho_t] \;=\; \int_\Omega d^3x \left(\tfrac{\kappa_c}{2}\,|\nabla R_{\text{coh}}|^2 + V(R_{\text{coh}}, \rho_t)\right), \qquad \kappa_c > 0. \tag{80}$$

A strictly positive coherence gap $\Delta_{\text{coh}} > 0$ follows if:

1. $R_{\text{coh}} = 0$ is a strict global minimizer of $V(\,\cdot\,, \rho_{t,0})$,
2. the Hessian at the vacuum is positive, $m^2_{\text{coh}} := \partial^2_R V(0, \rho_{t,0}) > 0$,

3. V is coercive for large R, e.g. $V(R, \rho_t) \geq c_4 R^4 - C$.

Under these conditions and the Poincare inequality on $\mathbb{T}^3_L$ (or suitable falloff on $\mathbb{R}^3$), one obtains the quadratic lower bound

$$E[R_{\text{coh}}] - E[0] \;\geq\; \int_\Omega d^3x \left(\tfrac{\kappa_c}{2} \, |\nabla R_{\text{coh}}|^2 + \tfrac{m^2_{\text{coh}}}{2} \, R^2_{\text{coh}} \right), \tag{81}$$

which implies a positive separation $E_1 - E_0 = \Delta_{\text{coh}} > 0$ between the vacuum and the first localized excitation.

Example C.1 (single-well quartic).

$$V(R, \rho_t) \;=\; a_2(\rho_t) \, R^2 + a_4(\rho_t) \, R^4, \qquad a_2(\rho_{t,0}) > 0, \quad a_4(\rho_t) \geq a_4^{\min} > 0. \tag{82}$$

Then $m^2_{\text{coh}} = 2a_2(\rho_{t,0}) > 0$ and coercivity is set by $a_4^{\min}$.

Example C.2 (convex polynomial).

$$V(R, \rho_t) \;=\; \sum_{k=1}^{K} b_{2k}(\rho_t) \, R^{2k}, \qquad b_2(\rho_{t,0}) > 0, \quad b_{2K}(\rho_t) \geq b_{2K}^{\min} > 0. \tag{83}$$

The smallest nonzero even coefficient fixes $m^2_{\text{coh}} = 2b_2(\rho_{t,0})$, while $b_{2K}^{\min}$ ensures coercivity.

Example C.3 (weakly ρ_t-dependent convex well).

$$V(R, \rho_t) \;=\; \alpha(\rho_t) \, R^2 + \beta(\rho_t) \, R^4, \quad \alpha'(\rho_{t,0}), \beta'(\rho_{t,0}) \text{ bounded}, \quad \alpha(\rho_{t,0}) > 0, \; \beta(\rho_t) \geq \beta_{\min} > 0. \tag{84}$$

Uniform bounds in a neighborhood of $\rho_{t,0}$ preserve $m^2_{\text{coh}} > 0$ and coercivity.

C.2 Uniformity in ρ_t and Stability of the Gap

Suppose ρ_t varies in a compact set $\mathcal{I}$ containing $\rho_{t,0}$ and the maps $\rho_t \mapsto a_2(\rho_t), a_4(\rho_t)$ (or the corresponding polynomial coefficients) are C^1-bounded on $\mathcal{I}$. Then there exist uniform constants

$$m^2_{\text{coh, min}} \;=\; \inf_{\rho_t \in \mathcal{I}} \partial_R^2 V(0, \rho_t) \;>\; 0, \qquad c_{4,\min} \;=\; \inf_{\rho_t \in \mathcal{I}} c_4(\rho_t) \;>\; 0, \tag{85}$$

and the bound (81) holds with m^2_{coh} replaced by $m^2_{\text{coh, min}}$. Consequently, Δ_{coh} is bounded away from zero uniformly for all admissible backgrounds $\rho_t \in \mathcal{I}$.

C.3 Defects, Winding, and Finite Energy

Although the vacuum sits at $R = 0$, localized excitations can support phase structure if $R > 0$ on a compact region while $R \to 0$ at its boundary, so that gradients of the phase do not produce long-range energy costs. A schematic construction on $\mathbb{R}^3$:

- Choose a compact domain $D \subset \mathbb{R}^3$ and a smooth profile $R(x)$ with $R(x) > 0$ for $x \in D$ and $R(x) = 0$ on ∂D and outside a thin collar of ∂D.
- On D, define a phase $\theta(x)$ with quantized winding along closed curves in D, while avoiding singularities by keeping $R > 0$ where $\nabla\theta$ is nonzero.

- Outside D (where $R = 0$), phase is irrelevant and does not contribute to energy.

In Abelian language, the circulation

$$\oint_{\gamma \subset D} \nabla\theta \cdot d\ell \;=\; 2\pi n, \qquad n \in \mathbb{Z}, \tag{86}$$

defines an integer winding that can be detected by loop holonomy when the gauge sector is present. Finite energy follows because R vanishes where θ would otherwise be singular, and $R \to 0$ outside D suppresses gradient and potential terms.

Vortex-like tubes and rings. In 3D one may realize line defects (tubes) or ring defects (closed loops) by taking D to be a solid torus and endowing the core with nontrivial winding. Energy remains finite provided $R \to 0$ sufficiently fast across the core boundary and at spatial infinity.

C.4 Stability Criteria: Energetic and Topological

Two complementary mechanisms support the existence and persistence of localized excitations:

1. **Energetic stability.** If $m^2_{\text{coh}} > 0$ and V is coercive, small perturbations of a localized solution increase energy (to quadratic order), preventing decay into the vacuum sector. This provides a local minimum of E constrained by localization.

2. **Topological stability.** If a configuration carries nontrivial winding on some compact support of $R > 0$, continuity and the vanishing of R on the boundary prevent unwinding without passing through $R = 0$ within D. When coupled to the gauge sector, such winding can be read out via holonomy. Topological charge is then an integer that cannot change under smooth variations preserving the support structure.

These criteria are compatible: energetic stability supplies a gap above the vacuum, while topology prevents decay channels that would bypass the energy barrier.

C.5 Illustrative Model Potentials

We list concrete choices useful in analysis and numerics.

C.5.1 Quartic with soft ρ_t modulation.

$$V(R, \rho_t) \;=\; \big(\alpha_0 + \alpha_1(\rho_t - \rho_{t,0})\big)\, R^2 \;+\; \big(\beta_0 + \beta_1(\rho_t - \rho_{t,0})\big)\, R^4, \tag{87}$$

with $\alpha_0 > 0$ and $\beta_0 > 0$, and $|\alpha_1|, |\beta_1|$ small. Then $m^2_{\text{coh}} = 2\alpha_0 + \mathcal{O}(\rho_t - \rho_{t,0})$ and coercivity persists.

C.5.2 Sextic stabilizer.

$$V(R, \rho_t) \;=\; a_2(\rho_t)R^2 + a_4(\rho_t)R^4 + a_6 R^6, \qquad a_2(\rho_{t,0}) > 0,\; a_6 > 0, \tag{88}$$

used when a_4 is small or mildly negative in some range; positivity of a_6 restores coercivity while $a_2(\rho_{t,0}) > 0$ secures the vacuum at $R = 0$.

C.5.3 Piecewise-smooth wells for compact support. For constructing compactly supported R, one may use a well with a steep wall near a target radius $R_\star$ inside D and a steep rise for $R > R_\star$, together with a strong penalty near the boundary to enforce $R \to 0$ across ∂D. Smoothing these features yields a C^2 potential that still satisfies the hypotheses above.

C.6 Consequences for Δ_{coh} and Observables

Under any of the examples above, one has $m_{\text{coh}}^2 > 0$ and coercivity, hence $\Delta_{\text{coh}} > 0$. In coupled SIT, loop holonomies measured over paths threading the support of a localized excitation detect phase winding and, indirectly, the presence of coherence through the dressing of gauge observables. Thus:

- Local measurements (e.g., two-point correlators of R) exhibit exponential clustering controlled by m_{coh} and by the first nonzero spatial mode on $\mathbb{T}_L^3$.
- Loop observables register nontrivial phases only when $R > 0$ along the enclosed support, consistent with coherence gating.

These features link the potential-level gap Δ_{coh} to experimentally or numerically accessible diagnostics.

Appendix D: Discrete-to-Continuum SIT

D.1 Graph Geometry and Variables

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a regular cubic graph embedded in the three-torus $\Omega = \mathbb{T}_L^3$ with lattice spacing $a = L/N$ and $|\mathcal{V}| = N^3$. For $i \in \mathcal{V}$ denote the spatial position by x_i, and for an oriented edge $(i \to j) \in \mathcal{E}$ write $x_{ij} = (x_i + x_j)/2$ and $\hat{e}_{ij} = (x_j - x_i)/a$. Discrete SIT variables:

$$R_i \approx R_{\text{coh}}(x_i), \qquad \rho_{t,i} \approx \rho_t(x_i), \tag{89}$$

$$U_{ij}^{(U(1))} = \exp\big(ia\, A_\mu(x_{ij})\, \hat{e}_{ij}^\mu\big), \qquad \mathcal{U}_{ij}^{(G)} = \exp\big(ia\, \mathcal{A}_\mu(x_{ij})\, \hat{e}_{ij}^\mu\big) \in G, \tag{90}$$

with $U_{ji} = (U_{ij})^{-1}$ and similarly for $\mathcal{U}$. Gauge-electric variables E_{ij} and $\mathcal{E}_{ij}$ live on links and satisfy discrete Gauss constraints.

D.2 Discrete Differential Operators

For a scalar f_i and link field A_{ij},

$$(\nabla_a f)_i := \frac{1}{a} \sum_{j \sim i} \alpha_{ij}\, (f_j - f_i), \qquad \alpha_{ij} = \frac{1}{6} \text{ on a cubic graph}, \tag{91}$$

$$(\Delta_a f)_i := \frac{1}{a^2} \sum_{j \sim i} (f_j - f_i), \tag{92}$$

so that $\Delta_a \to \Delta$ as $a \to 0$ for smooth fields. Magnetic field is encoded by plaquette holonomies. For an elementary square $\square = \{i \to j \to k \to l \to i\}$,

$$U_\square = U_{ij} U_{jk} U_{kl} U_{li} \;=\; \exp\big(ia^2 F_{\mu\nu}(x_\square)\, \hat{e}^\mu \wedge \hat{e}^\nu + O(a^3)\big), \tag{93}$$

$$\mathcal{U}_\square = \mathcal{U}_{ij} \mathcal{U}_{jk} \mathcal{U}_{kl} \mathcal{U}_{li} \;=\; \mathcal{P} \exp\big(ia^2 \mathcal{F}_{\mu\nu}(x_\square)\, \hat{e}^\mu \wedge \hat{e}^\nu + O(a^3)\big). \tag{94}$$

D.3 Discrete Hamiltonian and Gauss Law

A minimal nearest-neighbor Hamiltonian consistent with SIT reads

$$H_{\text{coh}}^{(a)} = \sum_i \Big[\tfrac{1}{2}\Pi_{R,i}^2 + \tfrac{\kappa_c^{(a)}}{2}\sum_{j\sim i}(R_j - R_i)^2 + V^{(a)}(R_i, \rho_{t,i})\Big], \tag{95}$$

$$H_{\text{gauge}}^{(a)}\big|_{U(1)} = \tfrac{1}{2}\sum_{\text{links}} E_{ij}^2 + \tfrac{\kappa_F^{(a)}}{2}\sum_{\text{plaquettes}} \big(1 - \operatorname{Re} U_\square\big), \tag{96}$$

$$H_{\text{gauge}}^{(a)}\big|_{G} = \tfrac{1}{2}\sum_{\text{links}} \operatorname{Tr}\mathcal{E}_{ij}^2 + \tfrac{\kappa_F^{(a)}}{2}\sum_{\text{plaquettes}} \Big(1 - \tfrac{1}{\dim R}\operatorname{Tr}_R \mathcal{U}_\square\Big), \tag{97}$$

$$H_{\text{int}}^{(a)} = \sum_i f^{(a)}(\rho_{t,i}, |\psi_i|)\, \mathcal{H}_{\text{SM}}^{(a)}(i) + \text{EFT corrections}. \tag{98}$$

Discrete Gauss constraints at node i:

$$\text{Abelian:} \quad \sum_{j\sim i} E_{ij} \;=\; \rho_i^{\text{matter}}, \tag{99}$$

$$\text{Non-Abelian:} \quad \sum_{j\sim i}\Big(\mathcal{E}_{ij} - \mathcal{U}_{ij}\mathcal{E}_{ji}\mathcal{U}_{ij}^{-1}\Big) \;=\; \mathcal{J}_i^0. \tag{100}$$

D.4 Coarse-Graining and Block Spins

Introduce a block scale $b \in \mathbb{N}$ and partition $\mathcal{V}$ into disjoint cubes of side ba. Define block-averaged variables

$$\bar{R}_I = \frac{1}{b^3}\sum_{i\in I} R_i, \qquad \bar{\rho}_{t,I} = \frac{1}{b^3}\sum_{i\in I}\rho_{t,i}, \tag{101}$$

$$\bar{U}_{IJ} = \operatorname{Proj}\Big(\prod_{(i\to j)\in\Gamma_{IJ}} U_{ij}\Big), \qquad \bar{\mathcal{U}}_{IJ} = \operatorname{Proj}\Big(\mathcal{P}\prod_{(i\to j)\in\Gamma_{IJ}} \mathcal{U}_{ij}\Big), \tag{102}$$

where Γ_{IJ} is a shortest-edge path between block centers and Proj returns the closest group element (for $U(1)$, the phase reduced modulo 2π). Iterating $a \to a' = ba$ generates a flow of couplings

$$(\kappa_c^{(a)}, \kappa_F^{(a)}, c_f^{(a)}, \ldots) \;\mapsto\; (\kappa_c^{(a')}, \kappa_F^{(a')}, c_f^{(a')}, \ldots), \tag{103}$$

which is used to match to continuum targets at a reference physical scale.

D.5 Practical Parameter Matching

Fix a physical matching scale μ_{match} (e.g., a small loop area and one lattice spacing for nearest-neighbor correlators). Choose tolerances $\varepsilon_{\text{loop}}$, ε_{coh}, ε_f and solve:

$$\Big|\langle 1 - \operatorname{Re} U_\square\rangle^{(a)} - \langle 1 - \operatorname{Re} U_\square\rangle_{\text{cont}}\Big| \le \varepsilon_{\text{loop}}, \tag{104}$$

$$\Big|\langle (R_j - R_i)^2\rangle^{(a)} - \langle (\partial_\mu R_{\text{coh}})^2\rangle_{\text{cont}}\Big| \le \varepsilon_{\text{coh}}, \tag{105}$$

$$\Big|\langle \mathcal{O}_{\text{gauge}}\rangle^{(a)} - c_f\,\langle |\psi|^2\rangle^{(a)}\Big| \le \varepsilon_f \quad \text{for small } |\psi|. \tag{106}$$

Tune $(\kappa_c^{(a)}, \kappa_F^{(a)}, c_f^{(a)})$ until all three conditions hold simultaneously. As a cross-check, verify that the effective mass gap extracted from temporal correlators satisfies

$$\Delta^{(a)} \;\ge\; \Delta_{\text{coh}}^{(a)} \quad \text{and} \quad \Delta^{(a)} \to \Delta_{\text{cont}} \;\; \text{within errors as} \;\; a \to 0. \tag{107}$$

D.6 Error Budgets and Scaling Windows

Discretization and finite-volume effects obey the canonical estimates on a smooth background:

$$\text{Discretization:} \quad \delta_{\text{disc}} \;=\; O(a^p), \quad p \geq 2 \text{ for central differences,} \tag{108}$$

$$\text{Finite volume:} \quad \delta_{\text{FV}} \;=\; O(e^{-L/\xi}) \;\; \text{for gapped spectra with } \xi = \Delta^{-1}. \tag{109}$$

A safe scaling window requires $a \ll \xi \ll L$ and stability of dimensionless composites (e.g., $\Delta\xi$, small-loop slopes, and ratios of correlator amplitudes). When performing block coarse-graining, keep b modest (e.g., 2 or 3) to avoid oversmoothing of small-loop diagnostics.

D.7 Continuum Consistency Checks

At fixed physical area A and separation r,

$$\langle 1 - \operatorname{Re} U_{\square}(A)\rangle^{(a)} \to c_A\, A^2 + O(A^{5/2}), \tag{110}$$

$$G^{(a)}(r) \to A_{\mathcal{O}}\, \frac{e^{-r/\xi}}{r^{\alpha}}, \tag{111}$$

with c_A proportional to $\langle F_{\mu\nu}^2\rangle$ (or its non-Abelian trace), and $\xi = \Delta^{-1}$. Agreement of these limits across several a and L establishes correct discrete-to-continuum matching of SIT and supports the persistence of the gap.

D.8 Summary

A regular cubic network with link holonomies, plaquette phases, and node-based coherence fields provides a faithful discretization of SIT. Coarse-graining defines a coupling flow used to match discrete parameters to continuum targets. Within a scaling window $a \ll \xi \ll L$, loop and correlator observables converge with controlled errors, and the discrete gap saturates the coherence floor predicted by the continuum theory.

Appendix E: Notation and Conventions

Spacetime and Metric

- Spacetime indices are Greek letters: $\mu, \nu, \rho, \sigma = 0, 1, 2, 3$.
- Spatial indices are Latin letters: $i, j, k = 1, 2, 3$.
- We use the mostly-minus Minkowski metric:

$$g_{\mu\nu} = \operatorname{diag}(+1, -1, -1, -1), \qquad g = \det(g_{\mu\nu}).$$

- For Euclidean calculations we use the flat metric $\delta_{\mu\nu}$ with all positive signs.
- Raising and lowering of indices is performed with $g_{\mu\nu}$ and its inverse $g^{\mu\nu}$.

Derivatives and Covariant Structures

- Partial derivatives: $\partial_\mu \equiv \frac{\partial}{\partial x^\mu}$.
- Covariant derivative acting on the coherence field ψ:

$$D_\mu \psi \equiv (\partial_\mu - iqA_\mu)\,\psi \quad \text{(Abelian)}, \qquad D_\mu \psi \equiv (\partial_\mu - ig\mathcal{A}_\mu)\,\psi \quad \text{(non-Abelian)}.$$

- Abelian field strength:

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- Non-Abelian field strength:

$$\mathcal{F}_{\mu\nu} \equiv \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - ig\,[\mathcal{A}_\mu, \mathcal{A}_\nu]\,.$$

- Holonomy (Wilson line) along a path γ:

$$U[\gamma] \equiv \mathcal{P}\exp\left(i\int_\gamma A_\mu\,dx^\mu\right) \quad \text{(Abelian)}, \qquad U[\gamma] \equiv \mathcal{P}\exp\left(i\int_\gamma \mathcal{A}_\mu\,dx^\mu\right) \quad \text{(non-Abelian)}.$$

  Here $\mathcal{P}$ denotes path ordering.

Gauge Groups and Generators

- For Abelian sectors we denote the gauge field by A_μ and the coupling by q.
- For non-Abelian sectors we take a compact simple group G with generators T^a in the chosen representation:

$$\left[T^a, T^b\right] = if^{abc}T^c, \qquad \mathrm{Tr}\left(T^aT^b\right) = \frac{1}{2}\delta^{ab}.$$

- The non-Abelian gauge field is $\mathcal{A}_\mu = \mathcal{A}^a_\mu T^a$ with coupling g.

Levi-Civita, Epsilon, and Contractions

- The Levi-Civita symbol is defined by $\epsilon^{0123} = +1$ and $\epsilon^{\mu\nu\rho\sigma}$ is totally antisymmetric.
- Contractions follow the metric signature; for example $X^\mu Y_\mu = g_{\mu\nu}X^\mu Y^\nu$.

Fourier and Integration Conventions

- Spacetime integration: $\int d^4x\sqrt{-g}\,\mathcal{L}$ in Minkowski space, $\int d^4x\,\mathcal{L}_E$ in Euclidean space.
- Spatial integration on a torus of side L: $\int_{[0,L]^3} d^3x$.
- A common Fourier convention is

$$\tilde{\phi}(p) = \int d^4x\, e^{-ip\cdot x}\phi(x), \qquad \phi(x) = \int \frac{d^4p}{(2\pi)^4}\, e^{+ip\cdot x}\tilde{\phi}(p).$$

Units and Dimensions

- Natural units are used unless stated otherwise: $\hbar = c = 1$.
- The gauge couplings q (Abelian) and g (non-Abelian) are dimensionless in $3+1$ dimensions.
- The coherence field ψ and its modulus $R_{\text{coh}} = |\psi|$ carry mass dimensions consistent with the kinetic term $|D\psi|^2$.

Sign Conventions

- The Lagrangian density for an Abelian gauge field is taken as

$$\mathcal{L}_{\text{U}(1)} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

- For non-Abelian fields:

$$\mathcal{L}_{\text{YM}} = -\frac{1}{2} \text{Tr}\left(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}\right).$$

- The kinetic term for the coherence field uses the mostly-minus metric:

$$\mathcal{L}_{\psi,\text{kin}} = \frac{\kappa_c}{2}\, g^{\mu\nu} (D_\mu \psi)^* (D_\nu \psi).$$

- The potential terms $V(R_{\text{coh}}, \rho_t)$ enter with a minus sign in the action:

$$\mathcal{L}_{\text{pot}} = -V(R_{\text{coh}}, \rho_t).$$

Miscellaneous

- Throughout, repeated indices are summed unless noted otherwise.
- "c.c." denotes complex conjugation; h.c. denotes Hermitian conjugation.
- Boldface symbols are not used for tensors; all indices are explicit.

Appendix F: Axiomatic Interfaces (Standard vs. SIT)

F.1 Why Wightman/OS Cannot Encode Coherence Gating

The Wightman and Osterwalder–Schrader (OS) axioms supply a mathematically precise framework for relativistic quantum field theory formulated in terms of operator-valued distributions (or Euclidean Schwinger functions) acting on a Hilbert space, with a fixed field algebra whose local observables are defined independently of the state being probed. In that setting, the gauge field is treated as a fundamental degree of freedom: gauge-invariant local operators (e.g. $\text{Tr}\, F_{\mu\nu} F^{\mu\nu}$) and Wilson-loop functionals are meaningful as elements of the observable algebra without requiring activation of an additional sector.

Coherence gating in Super Information Theory (SIT) is structurally different. In SIT, the gauge sector is not independently specifiable as a fundamental field algebra: gauge dynamics arise as phase geometry of the coherence fiber, and the renormalizable gauge/matter block is multiplied by a local dressing $f(\rho_t, |\psi|)$ that vanishes quadratically at the SIT vacuum. Consequently, "turning on" a physically observable gauge excitation is not a matter of inserting an operator built only from gauge

variables; it necessarily correlates with a nonzero coherence modulus and therefore with excitation of the coherence Hamiltonian. In particular, near the vacuum the relation $f(\rho_t, |\psi|) \geq c_f |\psi|^2$ enforces an energetic floor inherited from the coherence gap.

This mechanism is not representable within the standard Wightman/OS formulations without changing the axioms governing what counts as a local observable. If gauge-sector observables are dynamically nullified at the vacuum by a state-sensitive multiplicative dressing, then either (i) the would-be "gauge observables" are not elements of a fixed local *-algebra independent of the state, or (ii) one enlarges the fundamental field content and modifies the axioms so that the gauge sector is defined only conditionally (as an emergent, coherence-dependent subtheory). Either route is an axiomatic replacement rather than an instantiation of the standard program.

Accordingly, the SIT mass-gap theorem should be read as resolving the Yang–Mills mass-gap question *within the SIT axioms*, and simultaneously as a diagnosis: the standard axioms, by design, do not encode coherence-gated gauge accessibility, while SIT does.